

Factors influencing the export of commodities in Kenya

Diploma Thesis

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1 Introduction

1.1 Background

One of the main aspirations for the country's development policy is to increase and enhance the export capacity. Consequently, governments have put a lot of emphasis on the growth and promotion of exports. Economic literature shows that the economic growth and poverty alleviation measures are enabled by the export of commodities (Kiprono, 2019). Export promotion strategies undertaken by countries are in line with their comparative advantage; this requires that the country specialises in the production of goods which it can produce competitively. The country then avails those goods to the world at a cheaper price (Majeed and Ahmad, 2006). The income earned from exports is increased by enhancing the level of production, ensuring that the goods meet the required standards and the exploration of new markets (Hussain, Hussain and Alam, 2020).

The level of exports indicates the relative success or failure of the exporting entity. Most nations put in place measures and mechanisms to ensure that there is no disruption in the level of exports. The country focuses on the sustainability of exports because of the importance of exports on economic growth, sources of income and an avenue for competitiveness. The country can use the income from exporting commodities to achieve its potential. The performance of exports of commodities is critical for developing economies because it helps in the achievement of and sustenance of economic growth (Ge and Tang, 2020).

The export of commodities allows the developing nations to create links with the world economy, which increases the market base, increases the economies of scale and facilitates the transfer of technology (Sharma and Mishra, 2015). The export of commodities is important to developing nations because it is a key foreign exchange earner and links the country's consumption to its production (Bader, 2006). Daraisami (2004) argues that the new growth theories include exports as one

of the variables that bring about growth through various channels, such as an increase in capital equipment and the supply of raw materials. Thus, exports allow developing nations to enhance economic growth through imported capital goods.

Due to the tangible benefits that can be realised from the export of commodities and the development aspirations, Kenya has, over the last few decades, put in place measures to increase the export capacity, especially in the era of globalisation (Republic of Kenya, 2024). The government has put in place measures to increase the quantity and quality of commodities exported and to increase the foreign markets (Kiprono, 2019).

In most African countries, except for a few which are mineral resource-rich, the most important export commodities are agricultural commodities. In Sub-Saharan Africa, agricultural commodities export accounts for 40 per cent of the Gross Domestic Product (GDP) and approximately 15 per cent of exports. Further, more than 75 per cent of the population relies on work in the agricultural sector. This means that the agricultural sector is one of the most important sectors (Mugo, 2015). In Kenya, the most important export commodities include tea and coffee. In Kenya, the key export commodities are tea and coffee (Republic of Kenya, 2024).

Kenya is one of the main tea exporters globally. The sunny climate and the deep red soil provide the perfect conditions for the production of tea. In Africa, Kenya is one of the main tea producers. Thus, unsurprisingly, the production and export of tea contribute significantly to the economy of the country. Combined with coffee, spices and mate, they contribute approximately 20 per cent of the country's exports (Republic of Kenya, 2024). Figure 1 provides a summary of the tea production and exports in Kenya during the period 1980 – 2020. The trend analysis shows that the tea production has increased significantly from 120,000 tonnes in 1980 to 551,000 metric tonnes in 2020, which is a 359.17 per cent increase in production. Similarly, tea exports have increased from 75,000 metric tonnes in 1980 to 522,000 tonnes in 2020, which is an increase of 596 per cent increase. The increase in production and exports has been attributed to increased consumption due to the health benefits (Peluso and Monteiro, 2017; Shahidi, 2009). The increase in the production of tea is

three major factors. Firstly, the government has formulated policies to integrate small-scale growers into the main tea growing. Secondly, tea production in the country is done under efficient estate sectors by tea companies. This type of management has increased production. Third, under farmers are planting high-yielding varieties, improving cultivation methods and selective use of herbicides (Kenya Institute for Public Policy Research and Analysis (KIPPRA), 2020).



Fig. 1: Tea Production and Exports 1980 – 2020

Source: Tea Directorate of Kenya (2022)

Figure 2 shows the trend in coffee production and exports in Kenya during the period 1980 – 2020. The trend shows that coffee production declined from 91,000 tonnes in 1980 to approximately 38,000 tonnes in 2020, which represents a 58.24 per cent decline in the level of production. The decline in production is attributed to the prices of production, changes in rainfall and coffee prices (Republic of Kenya, 2024b). The low production levels are also attributed to low production levels; a tree produces 2 to 3 kilograms on average, as opposed to 30 kg. Additionally, the high cost of labour and the costs of fertilisers make it difficult for small-holder farmers to produce more coffee (Solidaridad, 2022).

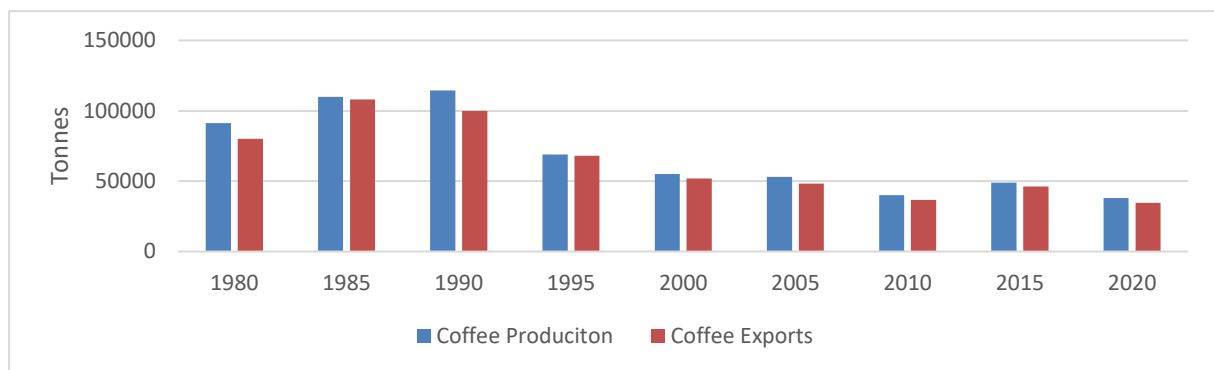


Fig. 2: Coffee Production and Exports 1980 – 2020

Source: Republic of Kenya (1980 – 2020)

1.2 Research Problem

Globalisation has provided economic integration by increasing foreign trade. International trade plays a significant role in the economic well-being of developing countries. Exports of commodities from Kenya play a significant role in the economic development of the country. Although the Kenyan government has an extensive export promotion strategy, providing a stable increase in exports has been subject to both internal and external shocks (Republic of Kenya, 2024). Numerous studies have been conducted to understand the factors that influence the export of commodities. The factors that influence the export performance of countries are categorised as either demand or supply-side factors. The demand side factors include the ability of trading partners to purchase the goods and services, the prices of exportable goods, the price of substitutes in the work market, the exchange rate, among other factors (Chand et al., 2022; Prakash and Maiti, 2016; Sami, 2020). The supply side factors include the productive capacity, the exchange rates, and the relative prices of competitive goods, the wage rate, and the costs of inputs (Handoyo, 2024). This research work hopes to continue to explore the factors that influence the export of commodities by focusing on Kenyan tea and coffee.

2 Objective

The main objective of the thesis was to determine, test and quantify the key factors that influence the export of commodities in Kenya with a focus on tea and coffee. Kenya exports numerous commodities, however, the research focused on tea and coffee exports.

2.1 Research Questions

- (i) What are the factors that influence the exports of tea in Kenya?
- (ii) What are the factors that influence the exports of coffee in Kenya?

This section provides theoretical and empirical perspectives of the effect of exports on exchange rates, economic growth and trade balance.

3 Literature Review

3.1 Kenya Export Promotion Policies

The objective of having export promotion strategies is to increase the trade stability, sustainability and the ability of the nation's goods and services to be competitive in the international market. The Kenyan government has, over the last few decades, developed strategies to increase the value of exports (Ngepah and Udeagha, 2018). Kenya has had five trade policy regimes. They include import substitution 1963-1979; Structural Adjustment Programmes 1980 – 1992; and Export-Oriented Policies. The import substitution policy was adopted via Sessional Paper No. 10 of 1965. This policy was adopted in the post-colonial era. The strategy aimed to develop, grow and protect the local industry. The objective of the policy was to reduce the value of imports and to substitute them with locally manufactured goods. This was achieved by reducing trade tariffs and quotas. The government also put in place measures to ensure that the locally manufactured products met international standards (Wanja, 2014; Republic of Kenya, 2017).

The import substitution policy did not achieve its intended objectives. In the 1980s, following years of economic challenges, the country adopted Structural Adjustment Programmes (SAPs). The objective of SAPs was to grow the economy. The focus shifted from import substitution and trade protectionism to export manufacturing and economic liberalisation. The policies focused on pushing the economy towards export promotion and economic reforms. The government set out to increase domestic efficiency and stimulate private sector developments. The government was seeking to increase foreign exchange earnings from manufactured goods (Republic of Kenya, 2017).

The SAP reforms were designed to provide a conducive environment for the liberalisation of the agricultural marketing systems. The policies included the production, pricing and marketing of domestic products and exports. For the manufacturing sector, the government developed bond schemes, duty-free, export guarantee

and credit schemes. During this period, the government founded the Export Promotion Council (EPC) and the export processing zones (EPZ). In 1981, Kenya joined the Preferential Trade Area (PTA) (Republic of Kenya, 2017).

Export-Oriented Policies (1993-2002) were aimed at increasing trade openness. This regime was part of the sixth development plan and policy of 1989-1993. During this period, the government developed the National Export Credit Guarantee Corporation (NECGC), reformulation and reduction of tariffs, development of export retention schemes and the removal of export duties and retention of export earnings. In 1995, Kenya joined the World Trade Organisation (Republic of Kenya, 2017). The strategy in economic recovery for employment and wealth creation (2003-2007) saw Kenya joining the EAC Customs Union.

Kenya Vision 2030 is the economic blueprint that is currently in operation in Kenya. The aim of the policy was to transform the economy into a middle-income country by the year 2030. Under this policy, the country joined the regional trade agencies such as the Africa Continental Trade Area (Afcfta), Tripartite Comesa-EAC-SADC –FTA. The country also joined trade partnerships with 10 countries.

3.2 Global Tea Trends

Tea production and consumption are believed to have started in China more than 2,000 years ago. Today, tea is one of the most widely consumed beverages across the globe. Global statistics indicate that after water, tea is the second most consumed substance (Meegahakumbura et al., 2018). In many developing nations, the tea sector contributes significantly to income and employment. Across the globe, tea production contributes approximately USD 17 billion per annum, while the trade in tea accounts for USD 9.5 billion per annum and thus tea is an important source of export earnings. One of the main features of tea production is that small-scale farmers are responsible for approximately 60 per cent of world production (Food and Agriculture Organisation, 2021).

3.3 Theories

Many economic theories of foreign trade have been developed in the past. They include absolute advantage theory, comparative advantage theory, the Heckscher-Ohlin theory, Marshall-Lerner condition, and the Single Equation Export model. These theories provide a framework for understanding the factors that influence the export of commodities. The theories provide a framework for analysing the relationship between the independent and dependent variables.

3.3.1 Absolute Advantage Theory

This theory is associated with Adam Smith. The concept of absolute advantage focuses on the ability of the individual, firm or country to produce goods and services using technologies in a manner that is more efficient than the competitors. Adam Smith introduced the concept of international trade. In that approach, Smith argued that absolute advantage in international trade is gained through labour productivity. In this theory, the productive capacity of the nation drives economic growth.

In this theory, it is advocated that a nation specialises in economic activities based on commodities in that it has an absolute advantage. The nation should import those goods that the trading partners have an absolute advantage in. Each country is encouraged to export the commodities that it can produce efficiently because the absolute labour required per unit is less than that needed by competitors (Appleyard and Field, 2014).

3.3.2 Comparative Advantage Theory

This theory is associated with the work of David Ricardo. The assumptions of the model include:

- (i) Each country has a finite endowment of resources, with each resource being identical
- (ii) The inputs in the production process are mobile between the different production processes within the country. This implies that the prices of inputs are the same for all producers.

- (iii) The factors of production are not externally mobile, they cannot move from one country to another.
- (iv) The labour theory of value is used in the model. The relative value of a commodity is based solely on its relative labour content.
- (v) The quality and quantity of technology are fixed by both countries and differ between them.
- (vi) The marginal costs of production are constant
- (vii) Both economies are operating at full employment
- (viii) The economies have perfect competition, no single consumer or producer is large enough to influence the market; all players are price takers.
- (ix) No government restrictions on economic activity
- (x) No transaction costs
- (xi) The analysis is based on two countries and two commodities(Appleyard and Field, 2014)

In theory, Ricardo argues a goods should be produced at a relatively lower opportunity cost or autarky prices, i.e. at a lower relative marginal cost prior to trade. In international trade, gains arise when nations with different factor endowments and technological progress produce goods based on the fact that they can produce at a cheaper price (Appleyard and Field, 2014).

3.3.3 Heckscher-Ohlin Model

This theory was developed by Eli Heckscher and Bertil Ohlin and builds on the work of David Ricardo. The Heckscher-Ohlin (HO) model states that a country exports products whose factors of production are abundant and cheap and imports the products whose factors of production are scarce (Appleyard and Field, 2014). A country has a relative advantage over the in the production of goods and services for which it has relatively large amounts of inputs. Goods that need locally available inputs are cheaper to produce than those that need inputs that are imported.

The advantage arises because countries that are endowed with factors have access to cheap inputs. The theory holds that the country will export the goods that are most abundant in its soil and import those that are scarce. The objective of the theory is to explain the dynamics of international trade that are observed in the global economy. The HO theory maintains that international trade is determined by the differences in the factor endowments. In theory, endowments are relative and not absolute. For example, the United States has vast arable land and thus exports agricultural products; on the other hand, China has a vast supply of labour and thus exports manufactured goods (Leamer, 1995).

3.3.4 Single-Equation Export Demand Models

This model provides a basic framework that is used in empirical studies dating back to the 1930s. It is a single equation that focuses on the demand side of the determinants of export while assuming that the supply side responds to infinite price elasticity. The supply side is assumed to exhibit idle productive capacity. The demand side is assumed to have increasing returns to scale (Nightrindi, 2008).

3.4 Price Fluctuations and Exports

According to Hallam (2003), the variability in the prices of commodities affects the supply and demand. A reduction in the prices of commodities is transmitted to the domestic market, which gives the producer the incentive to reduce production, but at the same time, the consumer gets an incentive to increase their demand for the product. According to Giordani, Rocha and Ruta (2012), small exporters take the prices of export commodities as given; thus, changes in the prices of commodities do not impact the global markets. However, because many small producers face the same international prices, there is an incentive to insulate them from fluctuations in prices, particularly in the domestic market. Thus, changes in the price of commodities have aggregate effects.

The international prices of food are often determined by the simultaneous imposition of export subsidies when prices are low and export taxes when prices are high. This simultaneously shifts the supply of the export commodity outward or inward, respectively. These policies have multiplier effects. When a shock causes the prices of export commodities to increase, exporters impose restrictions to mitigate the world price shock and to mitigate the chances of consumer loss aversion. All food-exporting countries impose restrictions when the prices increase. The government strives to maintain stable domestic prices (Giordani et al., 2012).

Unlike small exporters, large exporters do not take international prices on food as given. Food prices for large exporters are determined by their trade policy. In situations where the government determines that price increases could result in loss aversion, export policies are adjusted. Thus, when the large exporters increase the tax on exports, then the prices of the food item increase, which causes the exporting government to increase restrictions on exports to avoid consumer losses. To depress prices, the exporting country increases export subsidies, which reduces the prices and offsets the welfare loss of domestic food producers. For large exporters, commodity prices are strategic trade policies (Giordani et al., 2012).

Cunha et al. (2022) find that the price of trading partners' commodities impacts the export of a country's commodities. These were the findings of a study conducted to understand the effect of the prices of commodities exported by countries which imported manufacturing goods from Brazil. The research determined that there is a positive and direct link between exports from Brazil and the increasing prices of commodities trade by Brazil's trading partners. The increase in prices of commodities in the trading partner countries increased income, which in turn increased the purchasing power of the consumers.

In the intensive margin approach, the exports of a country are determined by the export prices relative to the competing prices. Therefore, increased foreign demand for domestic products is determined by prices, with lower prices stimulating foreign demand (Krugman, 1980). When using the autoregressive distributed lag

(ARDL) model and bound testing established that the export of commodities from Pakistan was determined by the prices (Hussain, Hussain & Alam, 2020).

Haniotis et al. (1988) found that the amount of export of wheat and soybeans is not affected by the prices of the commodities. The researcher found that the demand for the commodities was inelastic. However, the export of corn was determined by the price of the product. The demand for corn was found to be extremely elastic. Similarly, Abolagba et al. (2010) established that the export of commodities like cocoa and rubber from Nigeria was not determined by the price.

When studying how coffee prices in production and exports across the globe, Figueroa-Hernandez et al. (2019) found that the quantity of coffee produced across the globe is negatively related to the exports and positively associated with the price of Robusta and the London futures exchange. The researchers concluded that the international prices had the greatest impact on the variability in the world production and export of coffee.

3.5 Supply and Exports

The classical trade models, which include the Ricardo and Heckscher-Ohlin model, indicate that the supply of products to the global economy is determined by the comparative advantage of the economy. This comparative advantage comes about due to production levels. According to Kastrup and Kronborg (2021), international trade is due to the "love of variety". In the New Trade Theory, Krugman (1979), the economy's productive capacity determines the level of exports. The quantity and skills of the labour force determine the supply of commodities and how much the country can export. A large workforce enables the production of several different products.

The specification of an economy's export supply function is very complex because many factors determine the level of exports. The economic activities vary across the sectors and the commodities. However, there are two main conceptual approaches to model the determinants of export supply: the neoclassical approach

and the Keynesian approach. In the neoclassical approach, the export supply is determined by the relative prices and production capacity of the country. In the Keynesian approach, the export supply is determined by the domestic demand pressure (Sharma, 2003).

According to Gebreyesus (2015), the volume of exports of coffee and tea from East Africa is determined by the supply of the commodities. The supply is determined by the world prices and producer prices across the globe. However, domestic factors such as availability of credit, rainfall, the quality of soil and government policies determine the volume of tea and coffee that will be available for exports. According to Yoseph (2009), the elasticity of supply of coffee and tea is very high because the production levels are determined by the expectations of the producers.

According to Hussain et al. (2020), the level of exports in Pakistan is determined by the export supply. The export of manufactured and primary commodities such as rice, fruits and fish is determined by the production capacity and the domestic demand. Similarly, in China, Chan and Au (2007) established that an increase in production capacity increased the supply of textile exports. In 27 European Union Member countries, Sertić et al. (2015) established that an increase in the production of high-tech manufacturing capacity increased the amount of products available for exports.

Despite the findings of Kastrup and Kronborg (2021) and Hussain et al. (2020), some research studies find that the production capacity has a negative effect on the export performance. Jordaan and Eita (2012), when using annual data for the period 1999 to 2008 for 32 trading partners of South Africa found that the supply of leather and increases in the production capacity of leather did not have a significant effect on the export of leather products. In India, Sinha (2007) established that an increase in domestic production capacity did not increase the export performance of manufactured products.

According to the International Trade Administration (2024), the supply of coffee and tea from Kenya is determined by the use of pesticides and fertilisers. The use of fertilisers in the production of tea and coffee increased significantly during the

1990s due to the removal of government price controls and the import licensing quotas. The growth in usage occurred mostly among the smallholder farmers.

Maize was domesticated approximately 9,000 years ago. It has played an important role in the global food system. In previous years, the level of production and export of maize crops has increased. This has been attributed to technological advances, the increase in the output per hectare and expansion in the area under production. Maize is one of the main export cereals because of its high production volumes (Erenstein et al., 2022).

3.6 Other Determinants of Exports

Anagaw and Demissie (2013) found that in both the long and short run, export performance in Ethiopia was determined by infrastructure development, the level of financial development, the real exchange rate, trade openness, the real GDP of the home country and the trade openness.

In South Africa, Jordaan and Eita (2012) established that the determinants of leather exports from South Africa included the infrastructure of the importing country, the real effective exchange rate and some regional trade agreements. The growth in economic activities, indicated by an increase in the GDP, reduces the supply of leather available for export because growing income increases the domestic consumption of leather. Roy (2020) established that the determinants of the export of commodities from India were different. The export determination models indicated that agricultural exports of India were determined by several demand and supply-side factors. Lagged export production and world incomes were found to be the determining factors of India. The exports of rice and wheat, the production levels did not affect exports; rather, the exports of the products were determined by the reserves held by the government. The government policies determined the demand and supply of food commodities from India.

Kumar and Mittal (1995) found that tea exports are not affected by changes in the price levels and by changes in the global demand, and decrease with an increase in domestic consumption. Kumar (2010) established that the livestock exports from

India were determined by the ratio of world export to world output, exchange rate and the removal of quantitative restrictions positively, while relative prices did not significantly affect the level of livestock exports. In Uzbekistan, Halmurzaev (2015) established that the export of commodities such as cotton, metals, food and energy was determined by the levels of income in the trading partners and the internal monetary policies.

Sato (2020) studied the determinants of tea exports from Kenya using the Gravity Model Approach. The study focused on the 15 key destinations of tea exports. The study period was from 1990 – 2017. The researcher found that the main determinants of tea exports to the selected destination were the GDP of the importing nation, the value of the Kenyan shilling, the population, the distance, and having a common border. This research work continues the work of Sato (2020) by exploring the effect of other variables such as supply and price.

The study by Bamwesigye and Pomazalova (2015) established that the liberalisation of the coffee sector in Uganda has a positive and significant impact on the volume of coffee exports from the country. The liberalisation of the sector stimulated competitiveness, which increased the quality and quantity of coffee produced. Verter, Bamwesigye and Darkwah (2015) established that the coffee production and the world price index of coffee stimulate coffee exports from Uganda. The study also established that there was an inverse relationship between coffee exports and domestic coffee consumption.

According to Gylfason (1999), inflation is inversely correlated to the real exchange rate. High inflation rates reduce exports and export growth. Gylfason (1999) argues that inflation causes the overvaluation of the national currencies, which affects exports and economic growth. High levels of inflation also distort production by driving a wedge between the returns from investment in export production, which might reduce the level of export production. Ball and Sheridan (2005) established that inflation causes the prices of products and services to increase. This makes the cost of goods produced locally increase and makes them less competitive

internationally, thus reducing exports. When using the vector autoregressive approach on data from Nigeria, Okpe and Ikpesu (2021) established that inflation affected food exports negatively.

When studying the effect of inflation on exports in Indonesia, Malaysia, Philippines, Thailand and Vietnam during the period 2000 to 2015, Purusa and Istiqomah (2018) established that a one per cent increase in inflation level resulted in a 3.77 billion dollar reduction in exports. Akalpler (2013) established that in Turkey, high inflation rates during the period 1970 to 2000 or during the period of low inflation during the period 2000 to 2010 did not have any significant effect on the exports of the country. The researchers concluded that the trade capacity and trade policies were the key determinants of exports.

Fosu (1992), when studying the aggregate and individual agricultural export supply in Ghana, found that the real exchange rate (RER) did not have an impact on the country's agricultural exports directly; instead, the effect was due to incentives. The study focused on cocoa, coffee and sheanut. In the coffee model, it was established that the food price ratio and level of foreign income had significant but negative effects on coffee exports. A 10% increase in foreign income resulted in a 13.73% decline in the volume of coffee exports from Ghana.

3.7 Overview of the Tea Sector in Kenya

Tea was introduced to Kenya in 1903 by the British from India. During that period, the production and cultivation of tea was limited to multinationals and large-scale settlers (Patrick, 2013). After the country achieved independence in 1963, tea production and cultivation were opened up to all farmers. The liberalisation of the sector increased the area under cultivation and the production of tea. Figure 3 shows the trend in the production of tea and hectares under plantation. The trend analysis shows that the area under cultivation increased from 17,921 hectares in 1963 to 106,310 hectares in 2019 (Tea Directorate of Kenya, 2022). Tea is produced by large

estates and smallholder farmers. The trend in Figure 3 shows that most of the tea is produced by estates.

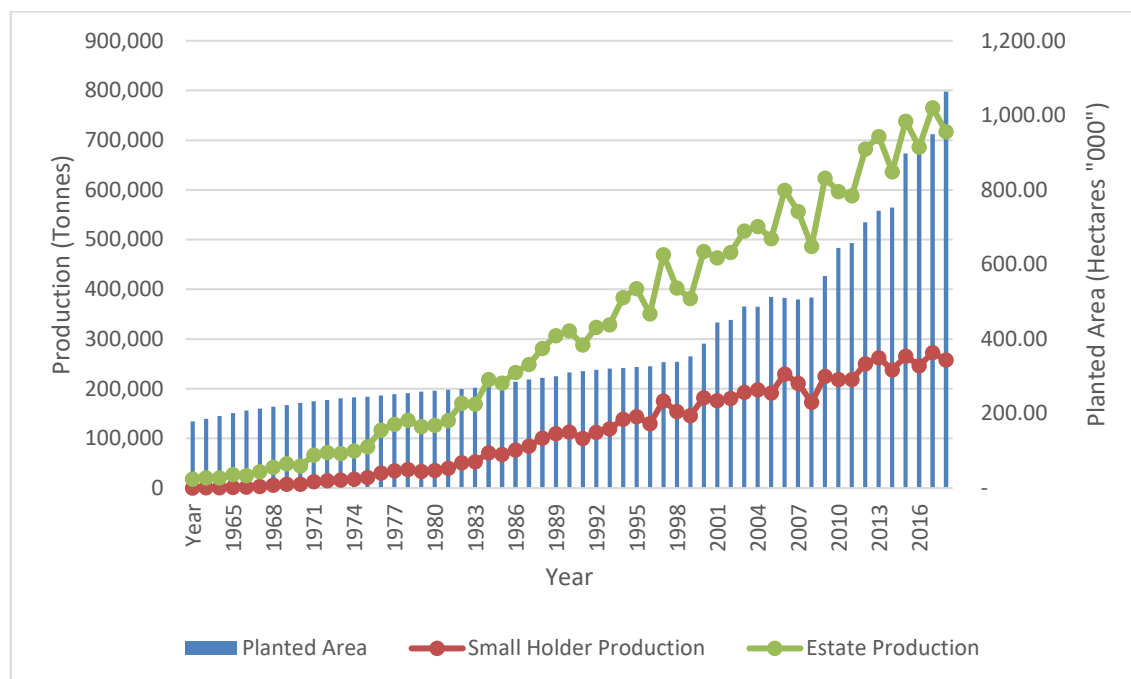


Fig. 3: Tea Production (Tonnes) and Planted Hectares (thousands in Kenya)
Source: Tea Directorate of Kenya (2022)

Figure 4 shows the trend in tea production, consumption and exports from Kenya. During the period 1963 to 2023, Kenya produced 13,994.357 metric tonnes (MT) and exported 13,638,547 (MT). Kenya exported approximately 97% of all tea produced during the period 1963 to 2023. During the same period, Kenyans consumed only 7% of the total tea production (Tea Board of Kenya, 2025). The trend shows that the level of tea exported is greater than the level of tea produced this is attributed to the fact that in some years, tea production exceeds exports and the surplus is sold in succeeding years additionally, approximately 10% of all tea exports is blended with tea from the East Africa region (East Africa Tea Traders Association, 2024).

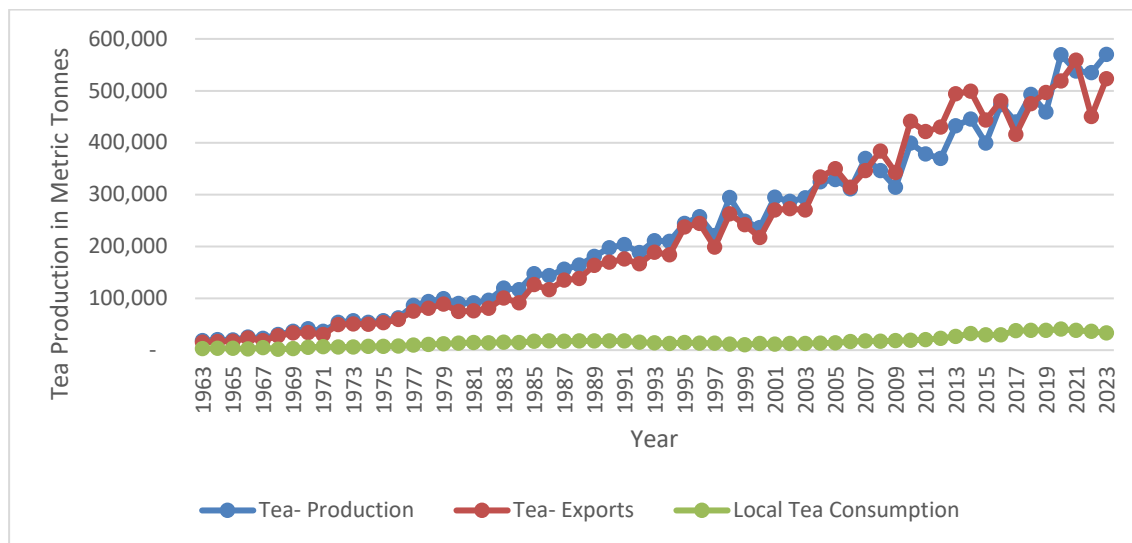


Fig. 4: Tea Production, Kenyan Consumption and Exports

Source: Tea Directorate of Kenya (2025)

Table 1 shows the top ten tea producers globally during the year 2023. The data indicates that Kenya is the third largest producer of tea globally after China and India. China produces approximately 40 per cent of the total tea output. India is the largest consumer of tea. Tea in Kenya is grown mostly in the highland areas (Times of India, 2024).

Tab 1: Top Ten Tea Producers 2023

Country	Tea produced (in tonnes)
1. China	2,400,200
2. India	900,144
3. Kenya	305,234
4. Sri Lanka	300,745
5. Turkey	175,090
6. Indonesia	157,855
7. Vietnam	117,001
8. Japan	89,496

9. Iran	84,321
10. Argentina	70,891

Source: Times of India (2024)

3.8 Overview of the Coffee Sector in Kenya

The area under coffee production by smallholder farmers increased from 49,121 hectares (ha) in 1963 to 117,677 ha in 1988. After this period, there was very little increment (stagnation) until 2008, when there was a significant decline. The area under cultivation declined from 122,040 ha to 84,263 ha in 2009. The area under coffee production by large-scale farmers remained fairly stable during the period under review, except for a sudden dip in 2008. The declining acreage under cultivation has reduced the level of production. Figure 5 shows the trends in coffee production by smallholders and estates. The trend shows that from 1978 to 2008, the amount of land under coffee for smallholders and estates increased. The increase was attributed to high prices, government loans to coffee farmers and subsidies for farmers. However, in 2009 and 2010, the area decreased sharply; thereafter, the level remained relatively stable, albeit at a lower level.

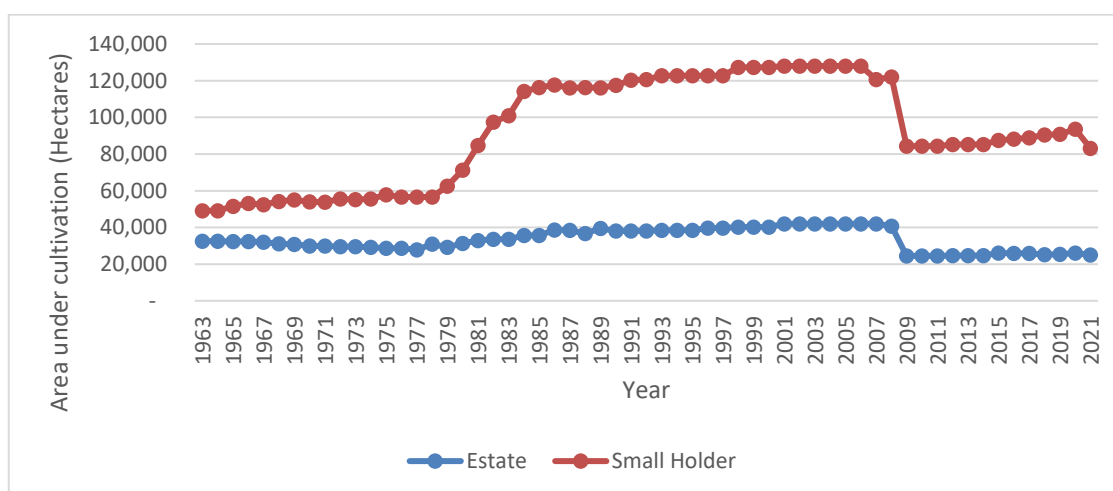


Fig. 5: Trends in Area Under Tea Cultivation

Source: Republic of Kenya (2024)

Figure 6 provides the trend in coffee production in Kenya during the period 1963 – 2021. The trend shows that the coffee production in the country increased stochastically during the period 1963 – 1987. This national trend was reflected in the estates as well as amongst the smallholder farmers. In 2001, there was a significant decline in the total production, production fell from 100,850 metric tonnes (MT) in 1999 to 50,543 MT in 2001. In 2021, the production level was 34,512 MT. The low production levels are attributed to falling rainfall in the coffee growing areas, lack of farm inputs and the change of land use from coffee farming to other enterprises (Republic of Kenya, 2024).

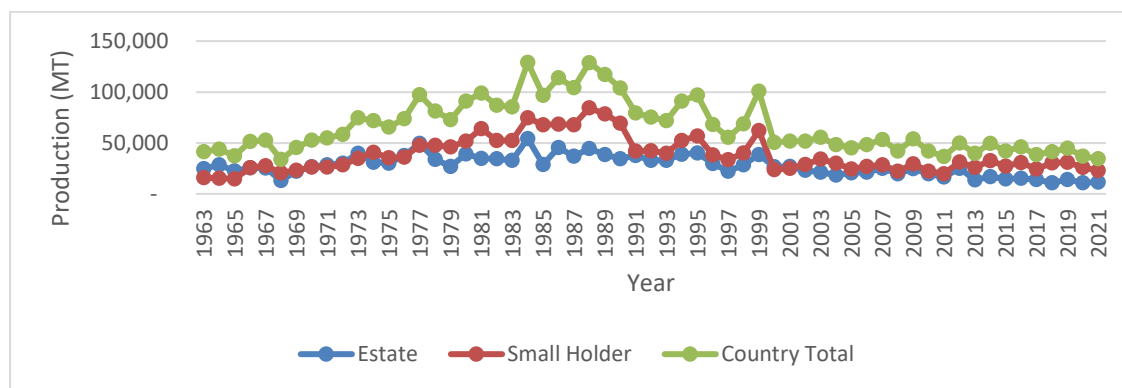


Fig. 6: Coffee Production in Kenya

Source: Republic of Kenya (2024)

Table 2 shows that local consumption increased from 509.9 MT in 2010 to 1,655 MT in 2021. This indicates that Kenyans are beginning to appreciate locally produced coffee. Additionally, the increase in tea consumption is due to government efforts to promote the consumption of local coffee blends. The government hopes to make drinking coffee part of the local culture. The number of local roasters has increased the quality and quantity of coffee available to the local market (Ambani, 2024).

Tab 2: Kenyan Coffee Consumption in Metric Tonnes

Year	MT	% Production
2010	509.9	1.21
2011	566.6	1.55
2012	629.5	1.26
2013	716	1.8
2014	744	1.5
2015	756	1.8
2016	771	1.67
2017	1050	2.75
2018	1342	3.24
2019	1411	3.14
2020	1577	4.27
2021	1,655	4.81

Source: Republic of Kenya (2024)

4 Materials and Methods

This chapter provides the details of the statistical method that was used to meet the objectives of the thesis. In particular, this chapter provides a description of the variables, the data source and the estimation technique. The study covers the period 1993- 2023.

4.1 Material

The study of the factors that determine the exports of commodities in Kenya was conducted using secondary data. The data was collected from numerous sources, which are available online. The data for the study were collected from the Coffee Board of Kenya, the Kenya Tea Directorate, the Central Bank of Kenya, and the World Bank database for commodities. The data for the study covered the period 1963-2023, the study covered 61 years. In 1963, the country gained full independence from Britain. The dependent variables of the study included tea exports (TE) and coffee exports (CE).

Table 3 provides the variables used in the tea model. In the tea model the independent variables in the study include own price, i.e. price of tea (PT) prices are given in USD per metric tonne; Income (INC) which is indicated using global GDP per capita (INC-Global) measured in USD (Kenyan tea is exported to most of the world); Area under tea production (TH), local tea consumption (LCT) in metric tonnes; and tea production (TP) in metric tonnes. The time series trends of the variables of the study are included in the Appendix.

Tab 3: Variables included in Tea Model

Variable	Type of Variable	Indicator	Source of Data	Expected Effect on Dependent Variable
Tea- Exports (TE)	Dependent	Demand for commodities	Kenya Tea Board	
Tea- Production (TP)	Independent	Output	Kenya Tea Board	Positive

Price Tea (PT)	Independent	Factors influencing commodity export/production	World Bank Commodities Prices	Positive/ Negative
GDP per capita (INC-Global)	Independent	Factors influencing commodity export/production	World Bank	Positive
Planted Hectares of Tea (TH)	Independent	Factors influencing commodity export/production	Tea Board of Kenya	Postive
Local Consumption of Tea (LCT)	Independent	Factors influencing commodity export	Tea Board of Kenya	Negative

Table 4 provides a summary of the variables used in the coffee model. The independent variables in the study include own price, i.e. price of coffee (PC) given in USD per metric tonne; Income (INC) which is indicated using GDP per capita from OECD (INC-OECD) countries for coffee exports (coffee exports from Kenya are consumed mostly from OECD countries); precipitation levels (PR) measured in millimetres per annum; and coffee production (CP) in metric tonnes.

Tab 4: Variables Included in Tea or Coffee Exports Models

Variable	Type of Variable	Indicator	Source of Data	Expected Effect on Dependent Variable
Coffee- Exports (CE)	Dependent	Demand for commodities	Coffee Board of Kenya	
Coffee – Production (CP)	Independent	Output	Coffee Board of Kenya	Positive
Price Coffee (PC)	Independent	Factors influencing commodity export/production	World Bank Commodities Prices	Positive/ Negative
GDP per capita (INC-OECD)	Independent	Factors influencing commodity export/production	World Bank	Positive
Precipitation (PR)	Independent	Factors influencing commodity export/production	World Bank Kenya Data	Positive

4.2 Methods

This subchapter provides a description of the estimation approach used to determine the factors that influence the export of commodities in Kenya. This section provides a description of the estimation approach and the assumptions that need to be fulfilled to conduct the regression.

Specification

The commodities of interest are coffee and tea. The thesis analysed the production and export of tea and coffee. The functional forms are given as:

$$TE_t = f(PT_t, TP_t, TH_t, INC\ Global_t, LCT_t) \dots\dots\dots (1)$$

$$CE_t = f(CP_t, PC_t, INC\ OECD_t, PR_t) \dots\dots\dots (2)$$

The subscript t (1,...,T) denotes the time (years). The linear forms of the equations are given as:

$$TE_t = a_0 + a_1PT_t + a_2TP_t + a_2TH_t + a_3\ INC\ Global_t + a_4LCT_t + u_t \dots\dots\dots (3)$$

$$CE_t = \beta_0 + \beta_1PC_t + \beta_2CP_t + \beta_3INC\ OECD_t + \beta_4PR_t + \epsilon_t \dots\dots\dots (4)$$

Econometric Approach

The regression approach for this thesis is the Autoregressive Distributed Lag Model (ARDL). In the model, both the dependent and independent variables are related not contemporaneously but across historical (lagged) values. Equation (6) provides a general ARDL ($p, q_1, \dots, q_k$)

$$y_t = \alpha_0 + \alpha_1t + \sum_{i=1}^p \psi_i y_{t-i} + \sum_{j=1}^k \sum_{l_j=0}^{q_j} \beta_j l_j x_{j,t-l_j} + \epsilon_t \dots\dots\dots (6)$$

Where y_t is the dependent variable; $x_i, \dots, x_k$ are the independent variables; α_0 is the constant term; $\alpha_1, \psi_t, \beta_j$, and l_j are the regression coefficients. Pesaran et al. (2001) argue that ARDL models are advantageous because they can manage cointegrated variables. The ARDL models for this thesis were

$$\begin{aligned} \Delta TE_t = & \alpha_0 + \sum_{i=1}^n \alpha_1 \Delta TE_{t-1} + \sum_{i=0}^n \alpha_2 \Delta PT_{t-1} + \sum_{i=0}^n \alpha_3 \Delta TH_{t-1} + \\ & \sum_{i=0}^n \alpha_4 \Delta INC Global_{t-1} + \sum_{i=0}^n \alpha_5 \Delta TP_{t-1} + \sum_{i=0}^n \alpha_6 \Delta LCT_{t-1} + d_1 PT_t + d_2 TH_t + \\ & d_3 INC GLOBAL_t + d_4 PT_t + d_5 LCT_t + u_t \dots\dots\dots (7) \end{aligned}$$

$$H_0: d_1 = d_2 = d_3 = d_4 = d_5 = 0$$

$$H_1: d_1 \neq d_2 \neq d_3 \neq d_4 \neq d_5 \neq 0$$

$$\begin{aligned} \Delta CE_t = & \beta_0 + \sum_{i=1}^n \beta_1 \Delta CE_{t-1} + \sum_{i=0}^n \beta_2 \Delta PC_{t-1} + \sum_{i=0}^n \beta_3 \Delta CP_{t-1} + \\ & \sum_{i=0}^n \beta_4 \Delta INC OECD_{t-1} + \sum_{i=0}^n \beta_5 \Delta PR_{t-1} + f_1 PC_t + f_2 CP_t + f_3 INC OECD_t + \\ & f_4 PR_t + \epsilon_t \dots\dots\dots (8) \end{aligned}$$

$$H_0: f_1 = f_2 = f_3 = f_4 = 0$$

$$H_1: f_1 \neq f_2 \neq f_3 \neq f_4 \neq 0$$

In Equations (7), and (8) Δ denotes the first difference, the variables $d_1, \dots, d_5$ and; $f_1, \dots, f_4$ denote the long run elasticities and, u_t and ϵ_t are the error terms for the respective equations. The null hypothesis (H_0) states that there is no long run relationship, while the alternative hypothesis (H_1) states that there is a long run relationship. In this process, the F-test is used to estimate the cointegration amongst the variables (Pesaran et al., 2001).

The presence of cointegration Equations (7) and (8) is estimated using Ordinary Least Squares (OLS) to determine the speed of adjustment of tea exports, coffee

exports, tea production and coffee production back to the equilibrium and to determine the short run and long-run coefficients. The Error Correction Model (ECM) is reparametrized into restricted form equations (7) and (8) as follows:

$$\Delta TE_t = a_0 + \sum_{i=1}^n a_1 \Delta TE_{t-1} + \sum_{i=0}^n a_2 \Delta PT_{t-1} + \sum_{i=0}^n a_3 \Delta TH_{t-1} + \sum_{i=0}^n a_4 \Delta IN\ GLOBAL_{t-1} + \sum_{i=0}^n a_5 \Delta TP_{t-1} + \sum_{i=0}^n a_6 \Delta LCT_{t-1} + \theta ECT_{t-1} + u_t \quad (9)$$

$$\Delta CE_t = \beta_0 + \sum_{i=1}^n \beta_1 \Delta CE_{t-1} + \sum_{i=0}^n \beta_2 \Delta PC_{t-1} + \sum_{i=0}^n \beta_3 \Delta CP_{t-1} + \sum_{i=0}^n \beta_4 \Delta INC\ OECD_{t-1} + \sum_{i=0}^n \beta_5 \Delta PR_{t-1} + \vartheta ECT_{t-1} + \epsilon_t \quad (10)$$

The coefficients of the error correction term, θ and ϑ in the equations (9) and (10) estimate the short-run speed of adjustments back to the long-run equilibrium condition.

Time Series Data

In econometric analysis, there are three types of data: time series, cross-sectional and pooled (i.e. combination of time series and cross-section) data. This thesis used time series data. The data analysed was collected at regular time intervals, annually. Time series data are commonly used in econometric analysis. However, when using time series data, the data must meet certain conditions (Gujarati and Porter, 2009).

- (i) The data needs to be stationary. The data is said to be stationary when the mean and the variance are constant over time and the covariance between two periods is determined only by the distance of the lag between the two periods and not at the time at which the value of the covariance was calculated (Gujarati and Porter, 2009).

Mean: $E(Y_t) = \mu$

$$\text{Variance: } \text{var}(Y_t) = E (Y_t - \mu)^2 = \sigma^2$$

$$\text{Covariance: } \gamma_k = E [(Y_t - \mu)(Y_{t+k} - \mu)]$$

When the study data is nonstationary, it is not possible to generalise it over different periods. Nonstationary data cannot be used for forecasting. Nonstationary data creates a unit root problem. Nonstationary data is made stationary by integrating the data by order d (Gujarati and Porter, 2009).

- (ii) The data should not have autocorrelation (Gujarati and Porter, 2009).
- (iii) When regressing time series variables on another time series variable(s), when the computed R^2 is very high (>0.9) even in situations where there is no meaningful relationship between the variables, this situation is an example of spurious or nonsense regression (Gujarati and Porter, 2009).

Classical Assumptions

According to Gujarati and Porter (2009), a regression model needs to meet certain conditions (assumptions) for the results to be considered correct.

- Assumption 1: Linear regression model: This requires that the parameters used in the estimation are linear, but it may or may not be linear in the variables (Gujarati and Porter, 2009).

$$Y_i = \beta_1 + \beta_2 X_i + \mu_i$$

- Assumption 2: X values are independent of the error terms, which requires that the regressor X is fixed in repeated samples. This means that the X variable(s) and the error term are independent, $\text{cov}(X_i, \mu_i) = 0$ (Gujarati and Porter, 2009).
- Assumption 3: Zero mean value of disturbance μ_i . For every X_i , the mean or the expected value of the random sample μ_i is =0

$$E(\mu_i|X_i) = 0 \text{ (Gujarati and Porter, 2009).}$$

- Assumption 4: Homoscedasticity or Constant Variance of μ_i . This means that the variance in the error term or the disturbance is the same regardless of the value of the independent variable

$$\begin{aligned} var(\mu_i) &= E[\mu - E(\mu_i|X_i)]^2 \\ &= \sigma^2 \text{ (Gujarati and Porter, 2009).} \end{aligned}$$

- Assumption 5: No autocorrelation between the disturbances: Given two independent variables X_i and X_j , ($i \neq j$), the correlation of any two μ_i and μ_j ($i \neq j$) = 0. This means that the observations are collected independently

$$\begin{aligned} cov(\mu_i, \mu_j | X_i, X_j) &= 0 \\ cov(\mu_i, \mu_j) &= 0, X \text{ is non-stochastic (Gujarati and Porter, 2009).} \end{aligned}$$

- Assumption 6: The number of observations n must be more than the number of independent variables in the model (Gujarati and Porter, 2009).
- Assumption 7: The Nature of X variables: The X values in a given sample cannot be the same. The $var(X)$ must be positive. The data should also not have outliers in the X variables (Gujarati and Porter, 2009).

5 Results

This section of the thesis provides the results of the data analysis to determine the factors that influence the export of commodities in Kenya.

5.1 Descriptive Statistics

Table 5 presents the description of the variables of the thesis. The data is annual, covering the period 1963 – 2023, which is 61 years. The data shows that during the period researched, the average annual tea production (TP) and tea exports (TE) were 229,415.70 and 223,582.70 metric tonnes. This indicates that most of the tea produced in Kenya is destined for the export market. The average coffee production (CP) was 65,033.66, with the exports (CE) being 65,495.00 tonnes. The Kenyan coffee exporters import blends from the region to mix with the local blends. The maximum price of coffee (CP) per metric tonne was USD 4,952.85, while the minimum price of coffee was USD 607.18. This volatility in prices has meant that farmers have reduced coffee production to protect their incomes (Republic of Kenya, 2024).

The maximum price of tea (TP) during the period under review was USD 2,970.56, with the minimum USD 719.60. The fluctuations in tea prices were not as wide as those for coffee. In the study, the global per capita income (INC-Global) was included in the study. Kenyan tea is consumed across the globe, thus, the demand is driven by global incomes. The average global per capita income was USD 5,328.13. Kenyan coffee is mostly consumed in OECD countries, where OECD per capita income (INC-OECD) was included in the study. The per capita income in the OECD during the study period was USD 19,849.78. The data shows that the average hectares under tea product (TH) was 116,723.40 thousand hectares. The average annual local tea consumption (LCT) was 16,223.06 thousand metric tonnes, which was approximately 7.07% of annual production. The annual average precipitation was 733.43mm, which is less than the 900 – 1200mm annual average needed for coffee production.

Tab 5: Descriptive Statistics of the Variables (T=61)

Variable	Mean	Std. dev.	Min	Max
TP	229,415.70	163,317.10	18,082.36	570,300.00
TE	223,582.70	169,497.90	15,353.64	558,920.00
CP	65,033.66	25,420.24	33,761.00	128,941.00
CE	65,495.00	23,318.68	37,504.10	126,498.00
PC	1,793.61	895.71	607.18	4,932.85
PT	1,713.82	597.98	719.60	2,970.56
INC-Global	5,328.13	3,862.34	524.35	13,169.60
INC-OECD	19,849.78	14,066.87	1,561.25	46,724.02
TH	116,723.40	64,861.98	21,448.00	269,430.00
LCT	16,223.06	10,069.29	1,924.243	40,300.54
PR	733.43	146.84	498.21	1,123.45

5.2 Stationarity Tests

The Augmented Dickey-Fuller (ADF) and the Phillips Perron (PP) tests were used to determine the stationarity test. The results of the stationarity tests are summarised in Table 6. The results show that the price of coffee (PC) and precipitation (PR) were stationary at levels, while the other variables were stationary at first difference.

Tab 6: Stationarity Test Results

Variable	ADF test		PP test		Stationarity
	I(0)	I(1)	I(0)	I(1)	
TP	0.208	-7.124***	1.564	-11.732***	In difference
TE	1.291	-9.128***	0.398	-14.43***	In difference

PT	-1.727	-7.124***	-2.265	- 11.732***	In difference
INC- GLOBAL	1.392	-5.705***	1.729	-6.782***	In difference
TH	0.102	-6.947***	0.44	-11.24***	In difference
LCT	-0.399	5.071***	-0.448	-6.94***	In difference
CE	-1.918	-9.459***	-1.860	-9.530***	In difference
CP	-1.836	-8.185***	-2.121	- 10.623***	In difference
PC	-2.703*		-2.747*		In level
INC-OECD	1.106	-5.381***	1.300	-6.745***	In difference
PR	3.560***		-5.570***		In level

***significant at 1%; **significant at 5%; *significant at 10%

5.3 Optimal Lag Selection

Before conducting the ARDL estimation, the optimal lag selection was done. The Maximum lag length was chosen using the Akaike Information Criterion (AIC), based on the number of regressors included in the model. The maximum lag was determined to be 2.

5.4 Bounds Test

Table 7 shows the results of the bounds test. The Bounds test (Pesaran et al., 2001) was done using the level values of the variables. The computed F-Statistic is greater than the upper bound (I(0)). This means that there is a long-run relationship between the variables. Thus, the ECM was estimated for both tea and coffee exports.

Tab 7: Results of the Bounds Test for Tea and Coffee Models

Dependent Variable	F- Statistic	10%		5%		2.5%		1%	
		I(0)	I(1)	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)
Tea	15.986	2.26	3.35	2.62	3.79	2.96	4.18	3.41	4.68
Coffee	8.601	2.45	3.52	2.86	4.01	3.25	4.49	3.74	5.06

5.5 Regression Results

This section provides the results of the ARDL estimations. The results are presented for coffee and tea.

Tea Export Results

Table 8 shows the results of the ARDL model. The lags for the variables were TE PT TP TH INC LCT, lags(1 0 2 2 1 2), which were determined by the minimum Akaike Information Criterion (AIC). The model summary shows that the model is significant at the 1% level (Prob>F = 0.0000), which implies that the model substantially explains the variation in the level of tea export with a good match. The $R^2 = 0.9958$ and Adjusted $R^2 = 0.9946$, which suggests that the variables in the model explain approximately 99.5% of the variation in the level of tea exports during the time taken into consideration. The lag value of tea exports (TE) is integrated into the model to capture the previous year's effects. At the 5% significance level, a 1% increase in the previous year's tea exports results in a 22.26% increase in the current year's tea exports. The price of tea (PT) has a negative and significant effect on tea exports at the 5% significance level. A unit increase in the price of tea results in an 11.76 unit decline in the level of tea exports.

In the current year, a 1% increase in the level of tea production (TP) results in a 79.50% increase in the level of tea exports ($p < 0.0001$). In the first and second lag, the level of tea production is negative and statistically significant at the 5% and 1% levels. The first lag of hectares under tea production (TH) has a negative effect on tea exports ($\beta = -0.9120$; $p = 0.006$). After the first lag, the effect is positive and statistically significant (first lag $\beta = 1.3928$; $p < 0.0001$; long-run $\beta = 0.4818$; $p = 0.030$).

At the first lag, a unit increase in the level of income results in a 27.5645 unit increase in the level of tea exports; the increase is statistically significant, $p = 0.0010$. Thereafter, the effect of income on tea exports is not statistically significant. At the second lag at the 10% significance level, a 1% increase in local tea consumption (LTC) results in a 192.43% decline in the level of tea exports. In the first and current year, the effect of LCT is not statistically significant.

Tab 8: Estimated Coefficients of the ARDL Model for Kenya Tea Exports

ardl	TE	PT	TP	TH	INC	LCT	lags(1 0 2 2 1 2)	aic
	TE							
		Coefficient		Std. Error		t		P> t
TE								
L1.		0.2226		0.0966		2.3000		0.0260
PT		-11.7577		5.2633		-2.2300		0.0310
TP								
-.		0.7950		0.0941		8.4500		0.0000
L1.		-0.3653		0.1072		-3.4100		0.0010
L2.		-0.4393		0.0817		-5.3700		0.0000
TH								
-.		0.4818		0.2153		2.2400		0.0300
L1.		-0.9120		0.3142		-2.9000		0.0060
L2.		1.3928		0.2594		5.3700		0.0000
INC-Global								
-.		-0.3798		7.2457		-0.0500		0.9580
L1.		27.5645		7.8783		3.5000		0.0010
LCT								
-.		-1.1504		1.2554		-0.9200		0.3640
L1.		-0.2341		1.4215		-0.1600		0.8700
L2.		-1.9243		0.9960		-1.9300		0.0600
_cons		-6767.2400		7159.7650		-0.9500		0.3500
Obs= 59; F(13, 45)= 821.62; Prob>F = 0.0000; R ² = 0.9958; Adjusted R ² = 0.9946								

Table 9 shows the regression conducted with the error correction term (ECT). The results show that all the variables except the first level of TH have a significant effect on tea exports at the 1%, 5% and 10% significance levels. In the long run, the price of tea and local consumption have a negative effect on tea exports at the 5% and 1% significance levels, respectively. In the long run, the hectares (TH) under tea production and the level of income (INC) have a positive and significant effect on tea exports at the 1% level. The speed of adjustment to equilibrium (ECM) when there is a shock in the previous period's equilibrium level is corrected for in

the current period. The ECM is significant at the 1% level, which confirms the long-run stability of the model. The ECM is -0.7774, indicating that the speed of adjustments to long-run equilibrium is approximately 77.74% annually.

Tab 9: Error Correction Model of Kenya Tea Exports

D.TE		Coefficient	Std. err.	t	P> t
ADJ					
	TE				
	L1.	-0.7774	0.0966	-8.0500	0.0000
LR					
	PT	-15.1239	6.9848	-2.1700	0.0360
	TP	-0.0124	0.1743	-0.0700	0.9440
	TH	1.2382	0.3657	3.3900	0.0010
	INC-				
	Global	34.9677	5.3093	6.5900	0.0000
	LCT	-4.2561	0.7702	-5.5300	0.0000
SR					
	TP				
	D1.	0.8046	0.1232	6.5300	0.0000
	LD.	0.4393	0.0817	5.3700	0.0000
	TH				
	D1.	-0.4808	0.3339	-1.4400	0.1570
	LD.	-1.3928	0.2594	-5.3700	0.0000
	INC-				
	Global				
	D1.	-27.5645	7.8783	-3.5000	0.0010
	LCT				
	D1.	2.1584	1.1913	1.8100	0.0770
	LD.	1.9243	0.9960	1.9300	0.0600
	_cons	-6767.2400	7159.7650	-0.9500	0.3500

Table 10 provides a summary of the model diagnostic tests. The p-values for the Ramsey Test, Breusch-Godfrey, White Test and Normality Test are greater than

the significance level of 5%. This implies that the residuals do not suffer from autocorrelation, are homoscedastic and normally distributed. The model describing the relationship between the dependent and independent variables is correctly defined.

Tab 10: Diagnostic Tests for Tea Exports Model

Test	Test Statistics	Degrees of Freedom	p-value
Ramsey Reset	0.0301	3, 42	0.9936
Breusch-Godfrey Test (lags=3)	6.1443	3	0.1048
White Test	59	58	0.4387
Normality Test	4.1621	2	0.1251
Cumulative sum test for parameter stability	0.6701		0.9479

The CUSUM Square test result (Figure 7) suggests that the model has long-run stability. The plot is within the critical band, implying that the model is stable.

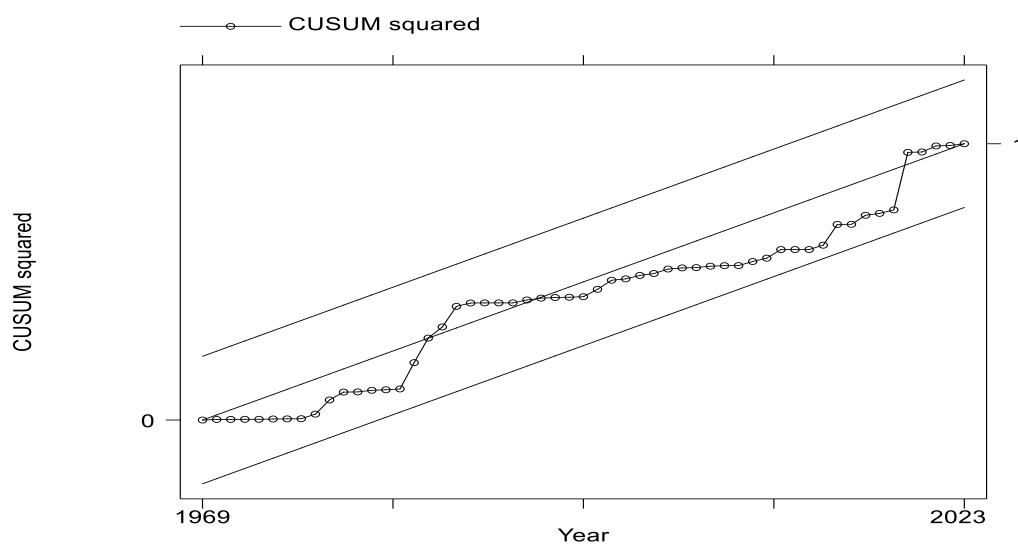


Fig. 7: Cusum Square Stability Test of Residuals from the ARDL Model of Tea Exports

The CUSUM test result (Figure 8) suggests that the model is stable in levels. The plot is within the critical band, implying that the model parameters are stable.

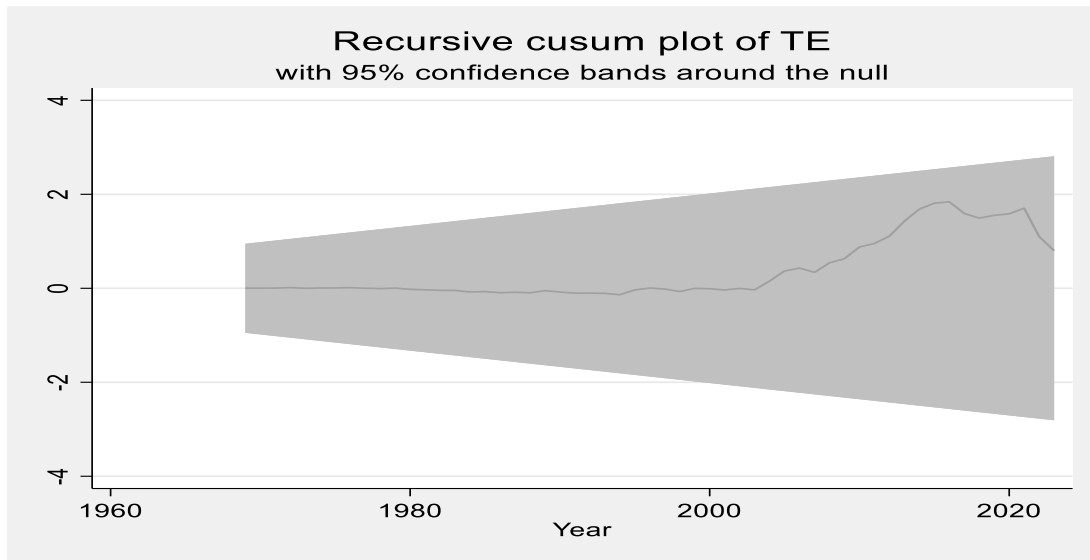


Fig. 8: Cusum Stability Test of Parameters from the ARDL Model of Tea Exports

Figure 9 shows the results of the Quantile-Quantile (Q-Q) plot. The points on the Q-Q plot fall roughly along the straight line, which implies that the data is normally distributed.

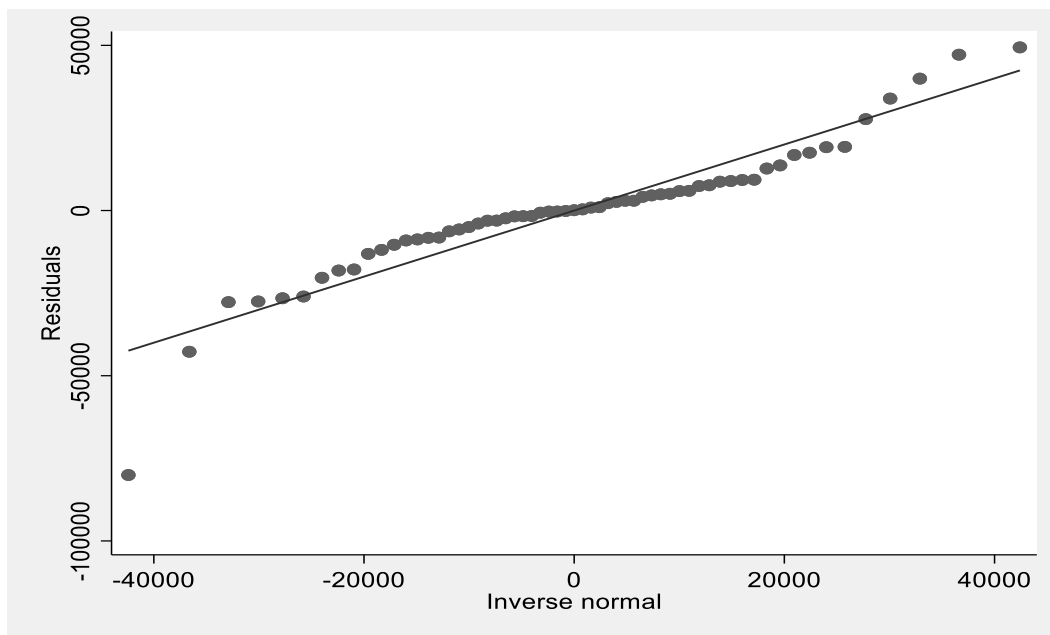


Fig. 9: Q-Q Plot of Residuals from the ARDL Model for Tea Exports

Figures 10 and 11 show the results of autocorrelation analysis using a correlogram. The results show that the data does not suffer from autocorrelation.

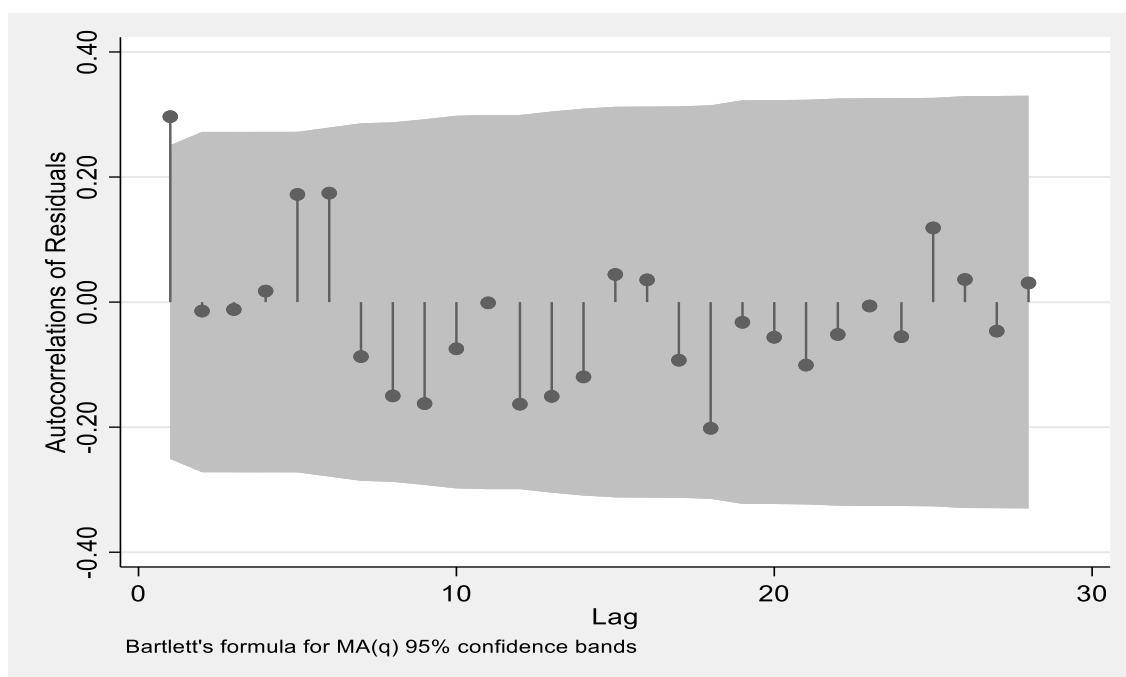


Fig. 10: Correlogram of Residuals from the ARDL Model of Tea Exports

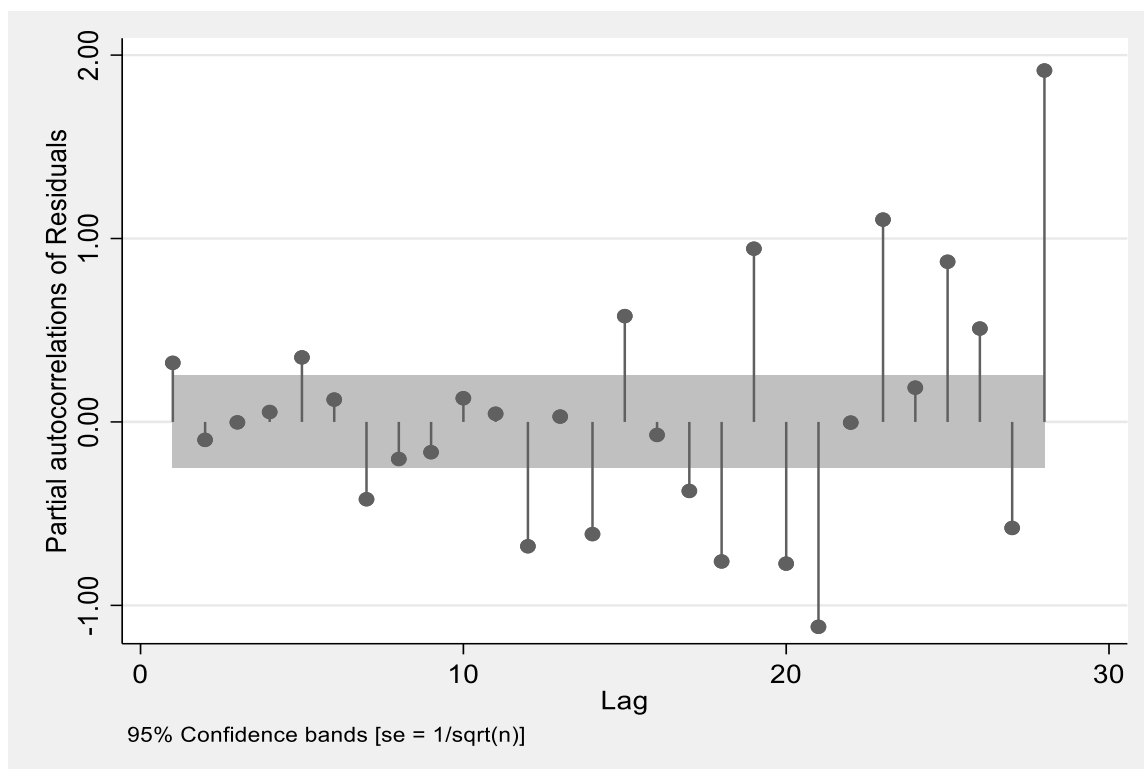


Fig. 11: Partial Correlogram of Residuals from the ARDL model for Tea Exports

Results from Coffee Exports Model

Table 11 shows the results of the ARDL model. The lags for the variables was CE, PC, CP, INC, PR, lags(1 0 2 0 0) determined by minimising the AIC criterion. The model summary shows that the model is significant at the 1% level (Prob>F = 0.0000), which implies that the model substantially explains the variation in the level of coffee export with a good match. The $R^2 = 0.8797$ and Adjusted $R^2 = 0.8632$, which suggests that the variables in the model explain approximately 86.32% of the variation in the level of coffee exports during the time taken into consideration. The lagged value of coffee exports (CE) is included in the model to capture the last year's effects. The results show that the previous period's coffee exports do not affect the current period's coffee exports ($p=0.241$). The results show that a unit increase in the price of coffee (PC) results in a 6.442 unit increase in coffee exports; the increase is statistically significant at the 1% significance level.

In the second lag, a unit increase in the level of coffee production (CP) increases current exports by 18.4%, which is statistically significant at the 5% level. A unit increase in the first lag of coffee production reduces coffee exports by 21.4%, which is statistically significant at the 10% level. A unit increase in the current level of CP increases exports by 53.5%, which is statistically significant at the 1% level. The results show that a unit increase in the level of income (OECD) and the precipitation levels reduces the level of coffee exports by 0.346 and -15.429 units, respectively, which is significant at the 1% and 10% levels.

Tab 11: Estimated Coefficients of the ARDL Model for Coffee Exports

ardl CE PC CP INC PR, lags(1 0 2 0 0), aic					
	CE	Coefficient	Std.err	t	p> t
CE					
	L1.	0.1737	0.1463	1.1900	0.2410
PC		6.4418	1.4867	4.3300	0.0000

CP				
-.	0.5349	0.1019	5.2500	0.0000
L1.	-0.2143	0.1148	-1.8700	0.0680
L2.	0.1841	0.0811	2.2700	0.0270
INC- OECD	-0.3457	0.0994	-3.4800	0.0010
PR	-15.4289	8.4495	-1.8300	0.0740
_cons	28133.0300	7921.7070	3.5500	0.0010
Obs= 59; F(7, 51)= 53.30; Prob>F = 0.0000; R^2 = 0.8797; Adjusted R^2 = 0.8632				

Table 12 summarises the results of the ECM model for coffee. The computed R-squared was 0.62, while the Adjusted R-squared was 0.5678. This implies that 56.78% of coffee exports during the period of the thesis were determined by the variables included in the model. In the short run, the first difference of coffee production has a significant effect on coffee export at the 5% significance level. In the long run, coffee production and the price of coffee significantly and positively increase the level of coffee exports at the 1% significance level. In the long run, increases in income (OECD) and precipitation have negative effects on coffee exports, which are statistically significant at the 1% and 10% significance levels, respectively.

The adjustment speed to equilibrium (ECM) when there is a shock from the previous period equilibrium, the error is corrected in the current year, is negative and statistically significant at the 1% level. This indicates that the model is stable in the long run. The ECM of -0.826 implies that the speed of adjustment to long-run equilibrium is approximately 82.5% annually.

Tab 12: Coffee Exports ECM Model

	D.CE	Coefficient	Std.err	t	P> t
ADJ					
	CE				
	L1.	-0.8263	0.1463	-5.6500	0.0000
LR					
	PC	7.7960	1.8159	4.2900	0.0000
	CP	0.6108	0.0744	8.2100	0.0000
	INC-OECD	-0.4184	0.1195	-3.5000	0.0010
	PR	-18.6723	10.5027	-1.7800	0.0810
SR					
	CP				
	D1.	0.0302	0.0994	0.3000	0.7630
	LD.	-0.1841	0.0811	-2.2700	0.0270
	_cons	28133.0300	7921.7070	3.5500	0.0010
R-Squared = 0.62; Adjusted R-Squared =0.5678					

Table 13 shows the results of the diagnostic tests. The model was evaluated for missing variables, serial correlation, heteroscedasticity and normality. The results show that the model was well fitted. The model did not suffer from serial correlation and heteroscedasticity. The model was normally distributed.

Tab 13: Diagnostic Tests for Coffee Exports

Test	Test Statistics	Degrees of Freedom	p-value
Ramsey Reset	1.6843	3, 48	0.1848
Breusch-Godfrey Test	6.0191	3	0.1107
White Test	43.9201	35	0.1433
Normality Test	2.1702	2	0.3371
Cumulative sum test for parameter stability	0.8167		0.9475

Figure 12 shows the results of the CUMSUMQ test, the results show that the variances are stable in the long run. The plot is within the band, which implies that the model is stable in variance.

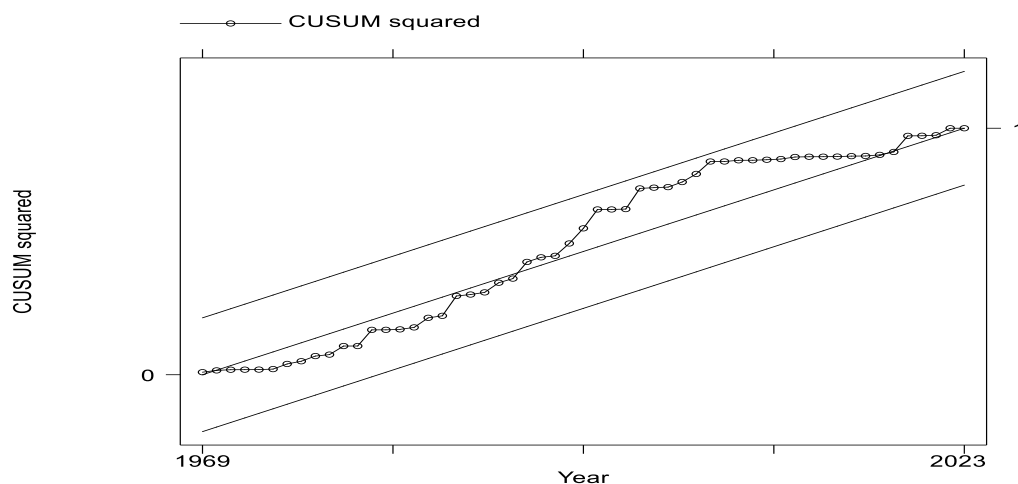


Fig. 12: Cum Results of Residuals from ARDL Model of Coffee Exports

The CUSUM test result (Figure 13) suggests that the model is stable in levels. The plot is within the critical band, implying that the model parameters are stable.

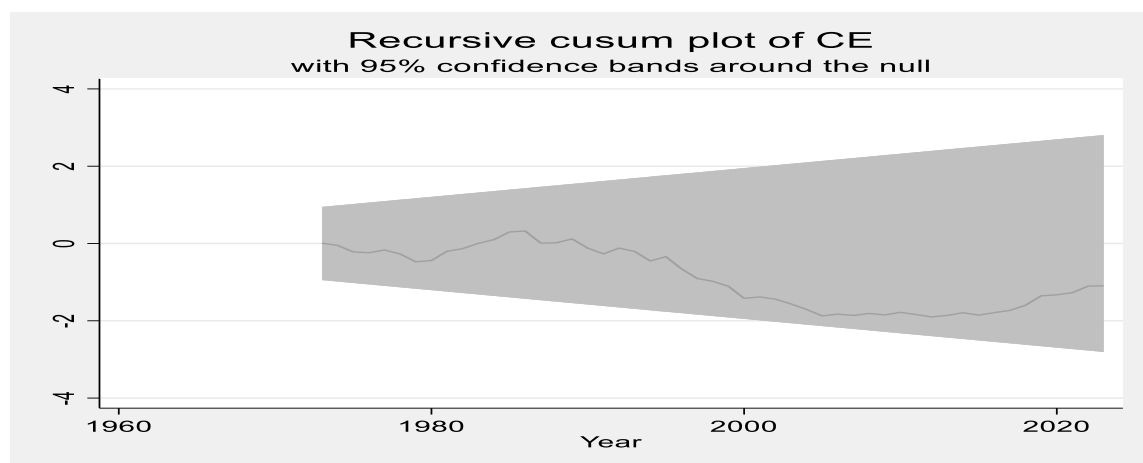


Fig. 13: Cusum Stability Test of Parameters from the ARDL Model of Coffee Exports

Figure 14 shows the results of the Quantile-Quantile (Q-Q) plot. The points on the Q-Q plot fall roughly along the straight line; this implies that the data is normally distributed.

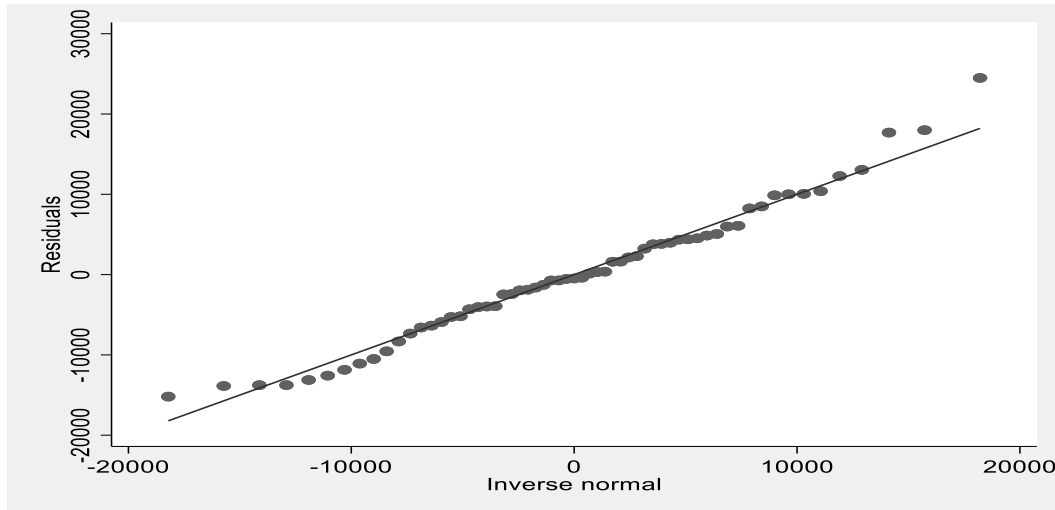


Fig. 14: Q-Q Plot of Residuals from the ARDL Model of Coffee Exports

Figures 15 and 16 summarise the results of the autocorrelation and partial autocorrelation tests. The findings show that the data did not suffer from correlation problems.

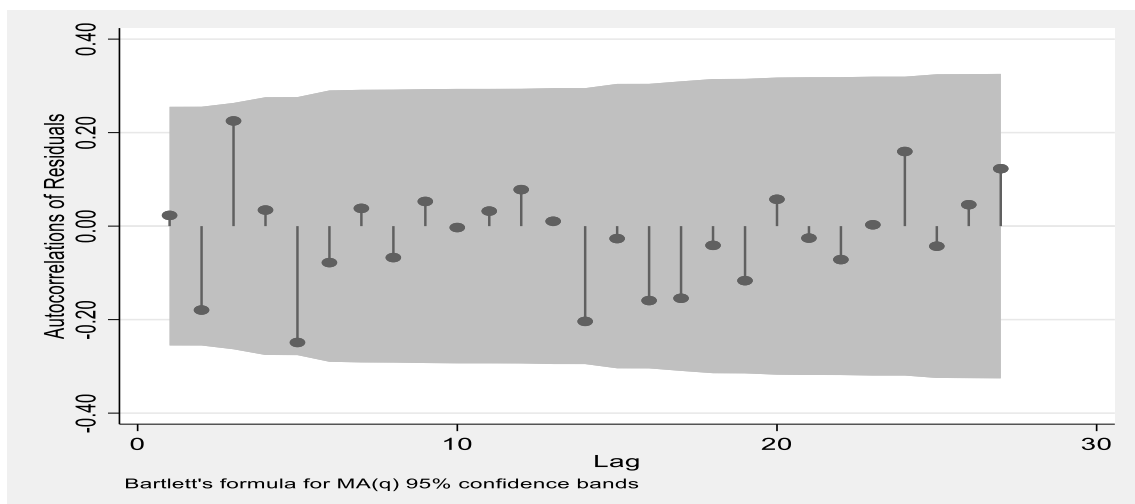


Fig. 15: Correlogram of Residuals from the ARDL Model for Coffee Exports

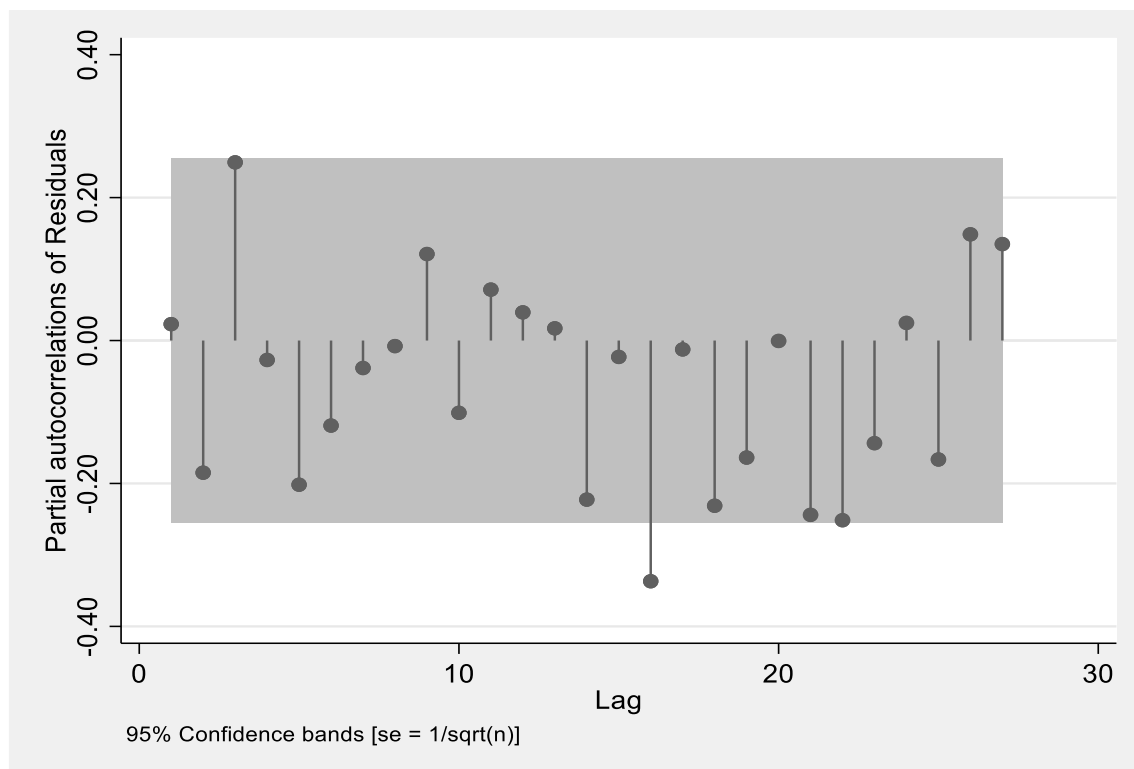


Fig. 16: Partial Correlogram of Residuals from the ARDL Model for Coffee Exports

6 Discussion of Results

This section of the thesis presents a discussion of the findings presented in Chapter 5. The results are discussed based on the models of the individual commodities.

Discussion of Tea Export Results

The results show that an increase in the previous year's export of tea increases the level of current year tea production. According to FAO (n.d.), an increase in the level of exports in the previous year stimulates the supply of the commodity in the current year. The previous year's performance signals increased profitability and stimulated the farmers to increase their level of output. High tea exports indicate that the demand is high.

The current price of tea was found to have a negative and significant effect on tea exports. Similarly, in the long run, the price of tea had a negative effect on exports. An increase in prices reduced the volume of exports. The model showed that there was no short-run relationship between tea and price. In economic theory, the laws of demand stipulate that an increase in prices results in a reduction. An increase in the price of tea reduces the export demand for the commodity. The findings contradict the findings of Cunha et al. (2022), who found that an increase in prices of manufactured commodities from Brazil increases the volume of exports. The results suggest that changes impact manufactured commodities and tea exports differently.

The empirical results show that an increase in the current year's level of tea production results in an increase in the current year's level of exports. This was expected. According to Hussain et al (2020), production level greatly enhances the supply of export supply. The results show that the first and second lag of tea production reduces the current year's level of tea exports. The findings suggest that tea production is not instantaneous; tea needs time to grow and to be harvested. The findings show that in the long run, the level of tea production on exports is insignificant. This is inconsistent with prior expectations. These results confirm the findings of Roy (2020), who established that in the long run in India, wheat and rice exports

are not affected by regulatory factors. The results show that the short changes in tea production affect output positively.

The results show that an increase in the area under tea production (tea hectares) results in an increase in the tea exports in the current year, first and second lag, in the long run and the first difference in the short run. These findings are in line with expectations. An increase in the area under production increases the volume of the commodity that is available for exports (Erenstein et al., 2022).

In economics, income has been theoretically and empirically shown to be a determinant of demand. The study by Halmurzaev (2015) showed that the level of exports of commodities is determined by the level of incomes in the trading partner. Kenyan tea is consumed across the globe, thus, global GDP per capita income was used as a proxy for income. The results of the regression show that the current year's income does not have a significant effect on tea exports, while the first lag of income has a positive effect on exports. In the long run, the effect of income is positive and significant, while in the short run, the effect is negative and significant. These findings partially confirm the findings of Halmurzaev (2015). The findings also confirm the findings of Fosu (1992) that in some instances, the export of commodities does not increase with an increase in income.

The results show that the second lag of local consumption of tea and the long-run consumption of tea have negative effects on tea exports from Kenya. The trend shown in Figure 4 indicates that local tea consumption in Kenya is very low, thus, the expectation was that the effect would be insignificant. However, local consumption reduces the amount available for exports. These findings confirm the findings of Kumar and Mittal (1995) and Verter et al. (2015).

Discussion of Coffee Exports Results

The regression results show that the previous period's level of coffee exports does not influence the current period's level of exports. The regression results show that an increase in the current period of prices of coffee and a long-run increase in coffee prices increase the level of coffee exports. These findings are similar to the findings

of Figueroa-Hernandez et al. (2019). The theories of supply show that the amount supplied is directly related to the prices; an increase in prices increases the level of supply. Overall, the findings show that increases in coffee production increase the level of exports in the current period, in the long run and the short run.

In the coffee model, GDP per capita for OECD countries was used as a proxy for income. The OECD countries consume a large percentage of coffee exports from Kenya. The regression estimation indicates that an increase in the level of income results in a reduction in the volume of coffee exports both in the current period and in the long run. This violates the theories of demand, which indicate that an increase in income results in an increase in consumption levels. The findings contradict the findings of Roy (2020), which showed that increases in world income increased the level of exports.

The regression results indicate that there is an inverse relationship between precipitation levels and rainfall levels. According to the Republic of Kenya (2024b), the changing rainfall patterns brought about by climate change are factors that contribute to the declining level of production of coffee and thus the level of exports.

Limitations

The thesis captures the impact of the price of tea, hectares under tea production, global income, tea production levels and local tea production on the value of tea exports. However, some of the variables have a bi-directional relation, for example, price affects production levels, and production levels affect the prices. Similarly, the coffee model included the impact of the price of coffee, coffee production, per capita income in OECD countries and precipitation levels. The variables have bi-directional relations, which were not captured in the model.

In Kenya, the coffee and tea industries are significantly affected by government policies and political interference. The actions of government and politicians significantly impact the price that the farmers receive. However, due to a lack of data or a proxy for these variables, they were not included in the study. In Kenya, tea and

coffee farmers receive payments for their products through government agencies. The payments are often delayed and do not reflect the international prices.

7 Conclusions

The main objective of this thesis was to determine the factors that influence the exports of commodities from Kenya. The commodities of interest in this thesis include coffee and tea. The thesis was based on the Absolute Advantage Theory, the Comparative Advantage Theory and the Heckscher-Ohlin Model. The thesis focused on the period 1963 – 2023. The data was collected from the World Bank, the Central Bank of Kenya, the Tea Directorate of Kenya and the Coffee Directorate of Kenya. The data was analysed using the Autoregressive Distributed Lagged approach. The variables in the study were found to be cointegrated, hence, the error correction model was estimated.

Based on the findings of the regression analysis, the thesis concludes that the current year's level of tea exports is increased by the previous period tea exports, current period tea production, current period and second lag of hectares under tea production, first lag of global income. In the current period, the level of tea exports in Kenya is negatively influenced by the current price of tea, the first and second lag of tea production, the first lag of hectares under tea, and the second lag of tea consumption. In the long run, tea exports are negatively influenced by the price of tea and local time consumption and positively influenced by hectares under production and global income. In the short run, tea exports are positively influenced by the first and second lag of tea production and local tea consumption and negatively influenced by the first difference of hectares under tea production and the first difference of global income levels. The speed of adjustment of tea exports to long-run equilibrium is -0.7774.

Based on the findings, the thesis concludes that the current period export of coffee is positively determined by the price of coffee, the current period and the second lag of coffee production, and negatively determined by income in the OECD countries and precipitation. In the long run, coffee exports are positively influenced by the price of coffee and coffee production and negatively influenced by income in

OECD countries and precipitation. In the short run, coffee exports are negatively influenced by the second lag of coffee production. The speed of adjustment of coffee to the long-run equilibrium is -0.826.

Recommendations

The purpose of this thesis was to understand the factors that influence the export of commodities in Kenya. The main commodities exported from Kenya include tea and coffee. In the thesis, two models were estimated with different independent variables. The tea export model was estimated using the price of tea, tea production, hectares under tea, income indicated by global GDP per capita and local tea consumption. The coffee export model had coffee price, coffee production levels, income in the OECD countries, which are the largest consumers of Kenyan coffee and the level of precipitation. In light of the findings summarised in Chapter 5, it is recommended that:

- (i) To improve the export capacity of tea and coffee in Kenya, the government should increase the hectares under production. Towards this, the government should provide loans for land acquisitions.
- (ii) To increase the productivity of coffee, the farmers need to take measures to deal with changing rainfall patterns. The government should encourage farmers to grow varieties of coffee that can withstand high rainfall levels.
- (iii) The price of tea fluctuates significantly, which affects the level of income for the farmers. The government should consider introducing a stabilisation measure to ensure that the farmers have a stable income. Export prices have a positive effect on coffee, however, the level of coffee production has been declining. Governments should ensure that the farmers get their payments on time and are compensated at minimum prices.
- (iv) Tea and coffee exporters in Kenya need to expand their tea markets to regions where income increases stimulate an increase in the level of tea

and coffee exports to ensure that they maximise the benefits of international trade.

- (v) The government must address supply-side constraints that have resulted in a decline in the hectares under coffee production. Coffee is a major income earner for the country, and increasing the area under production will increase income.

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Attachments

A Time Series Trends of Study Variables

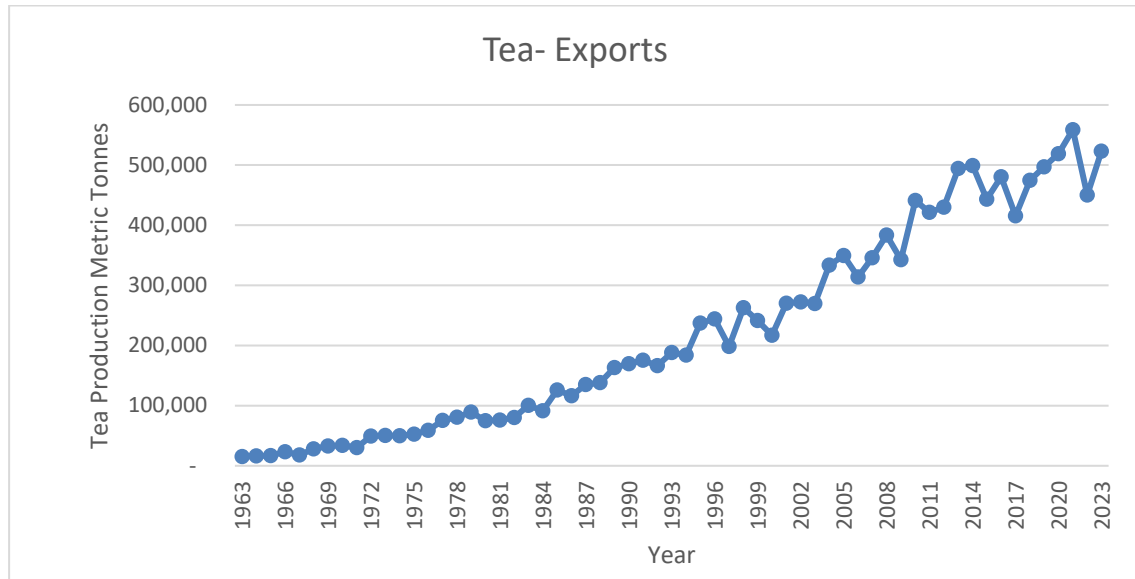


Fig. 17: Tea Exports Kenya , 1963 - 2023

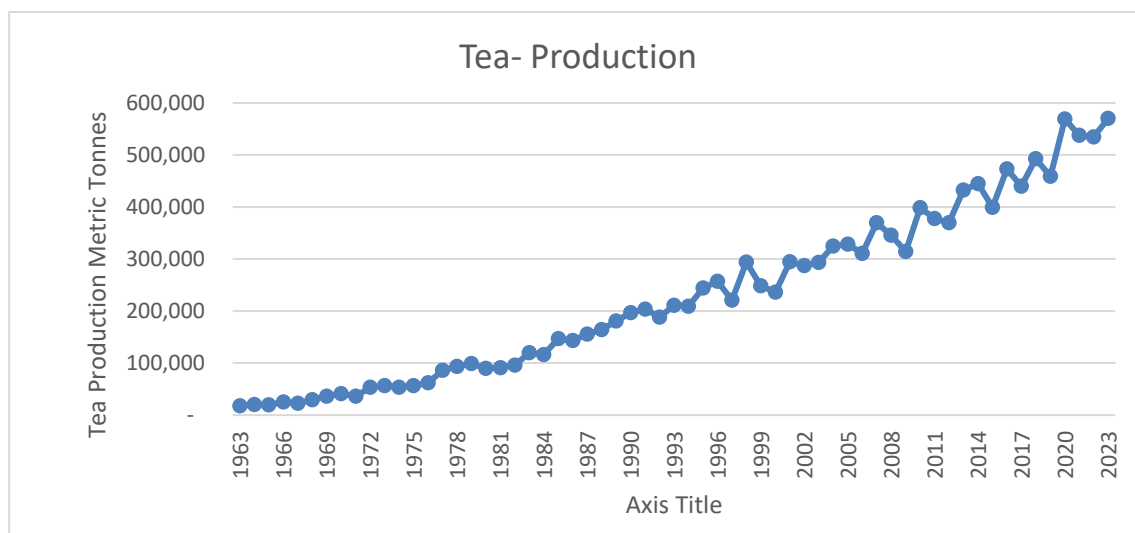


Fig. 18: Tea Production Kenya 1963 – 2023



Fig. 19: Price of Tea, 1963 - 2023

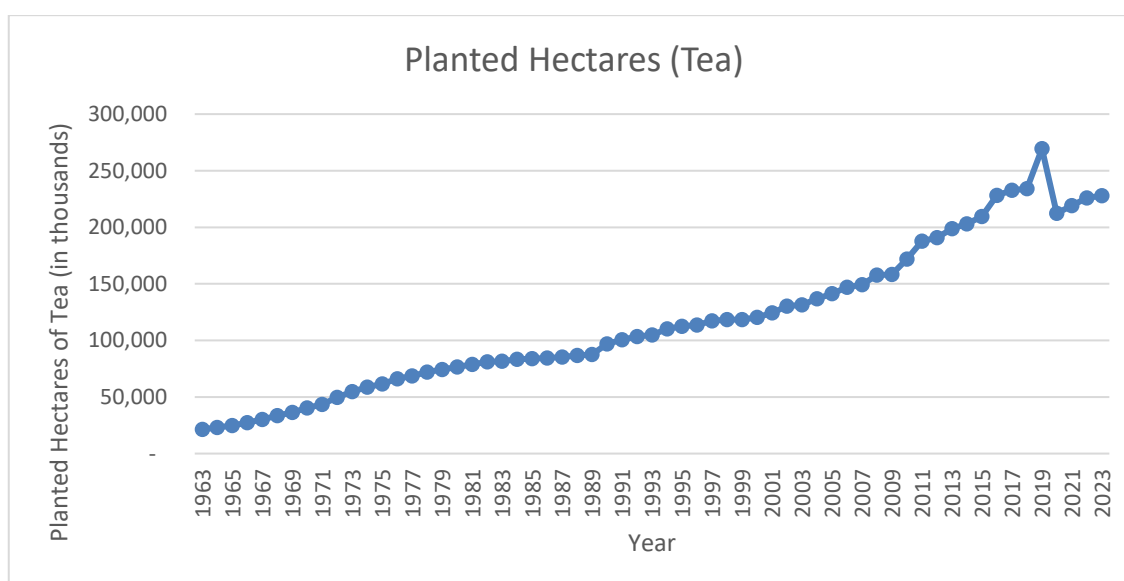


Fig. 20: Planted Hectares of Tea (in thousands) Kenya, 1963 - 2023

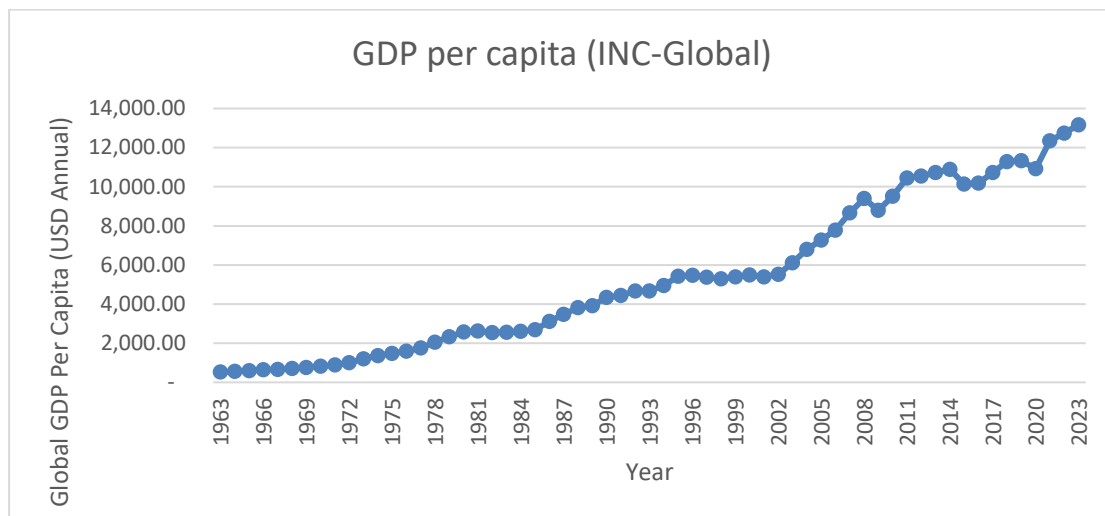


Fig. 21: Global GDP Per Capita (USD), 1963 - 2023

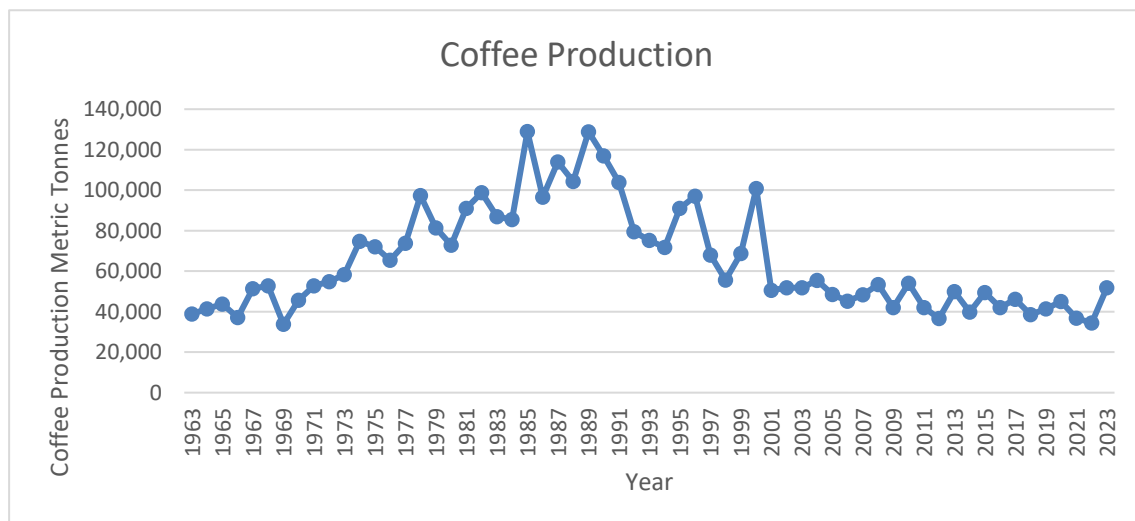


Fig. 22: Coffee Production in Kenya, 1963 - 2023

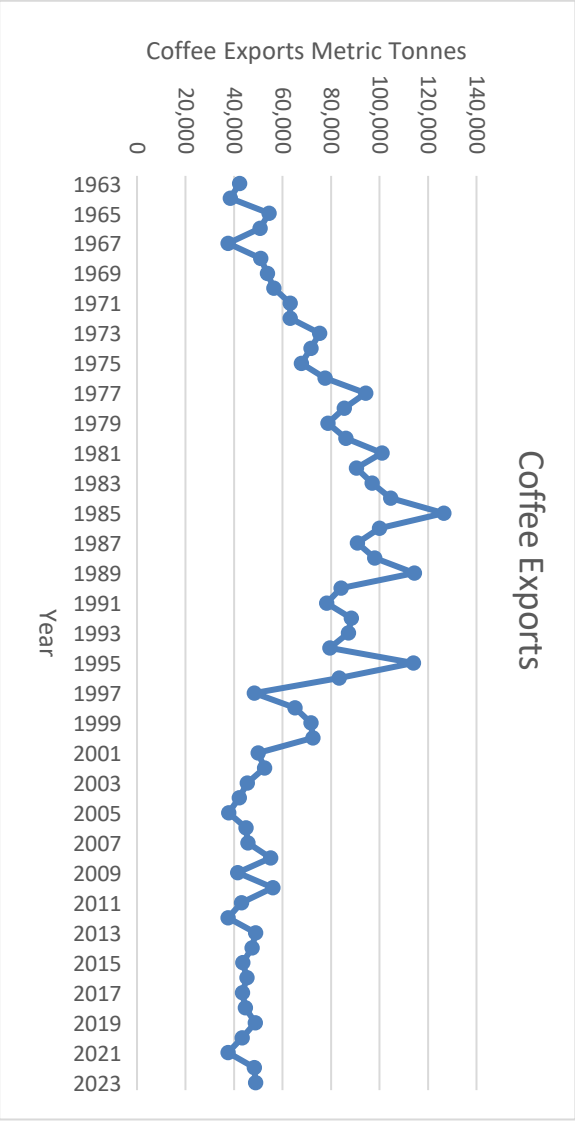


Fig. 23: Coffee Exports Kenya, 1963- 2023

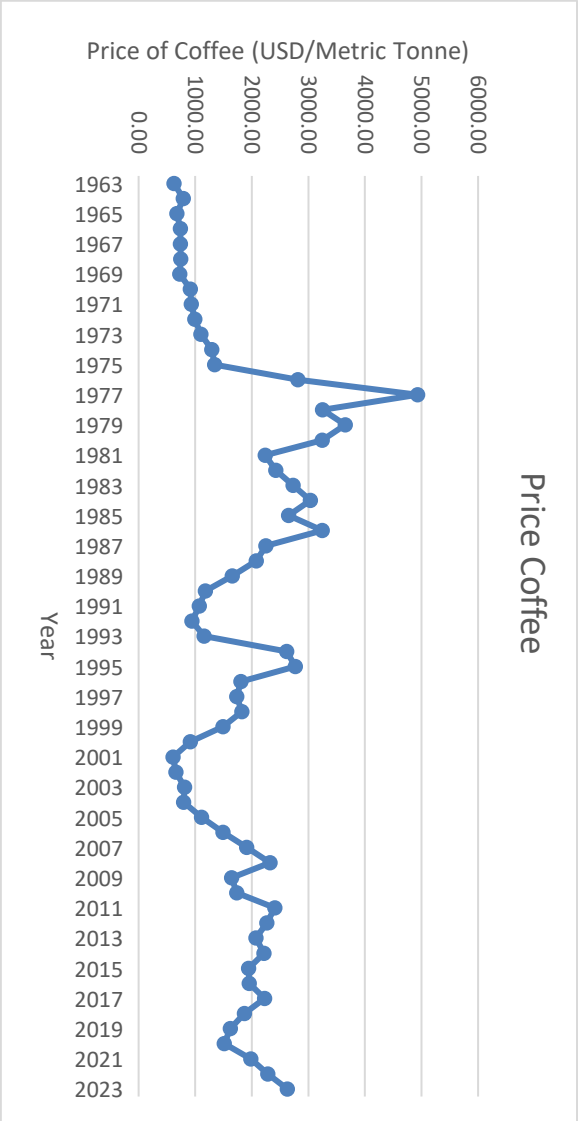


Fig. 24: Price of Coffee 1963 – 2023

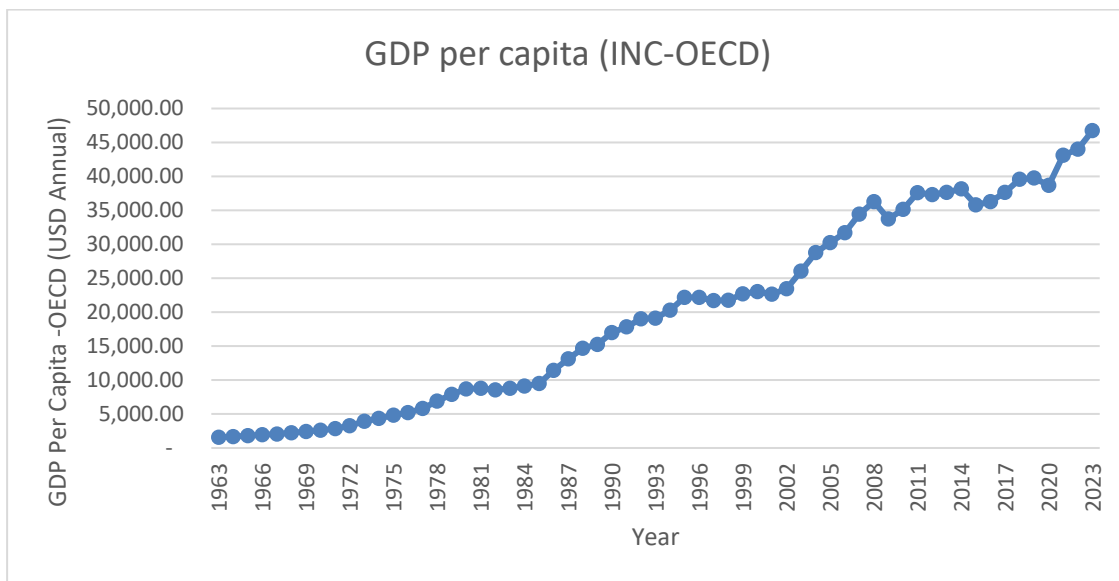


Fig. 25: GDP Per Capita in OECD, 1963 – 2023

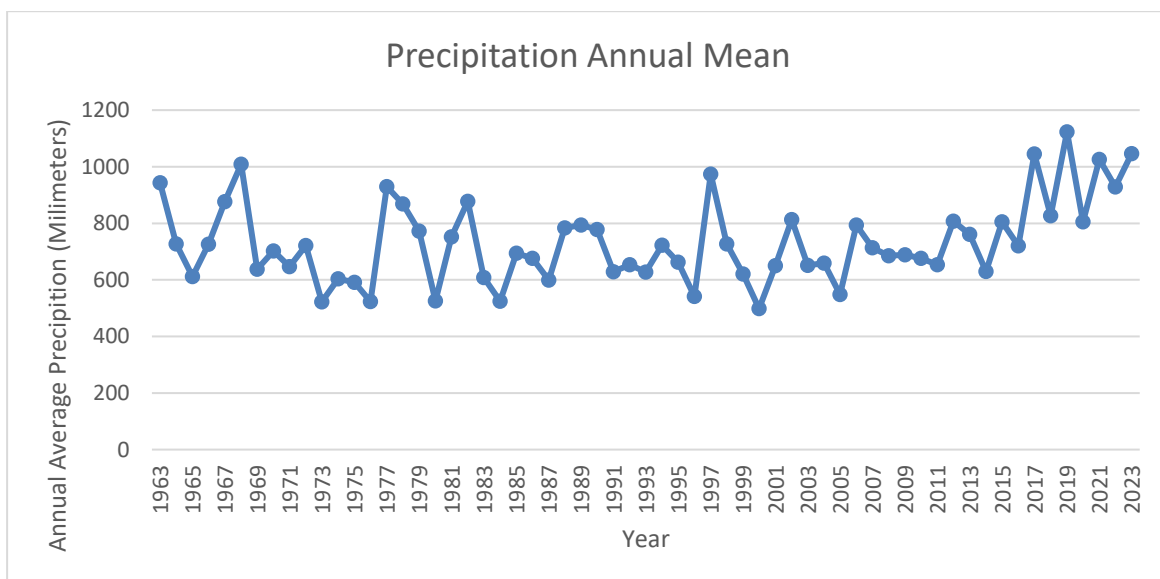


Fig. 26: Annual Average Precipitation Kenya, 1963 - 2023

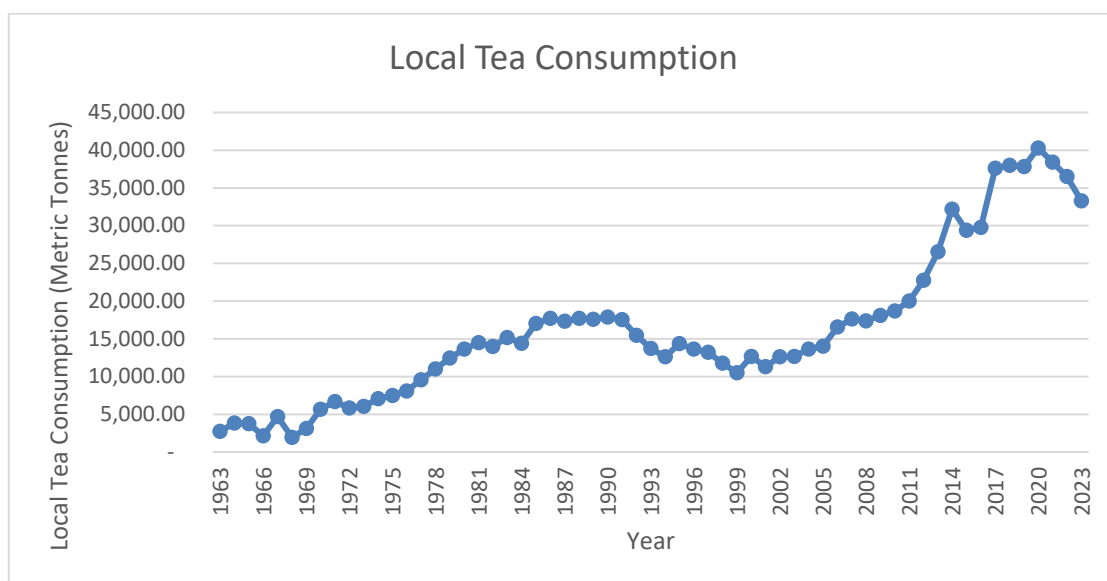


Fig. 27: Local Tea Consumption (Metric Tonnes) K